Structural Identification

(I) Hydrogenation of alkenes & alkynes.

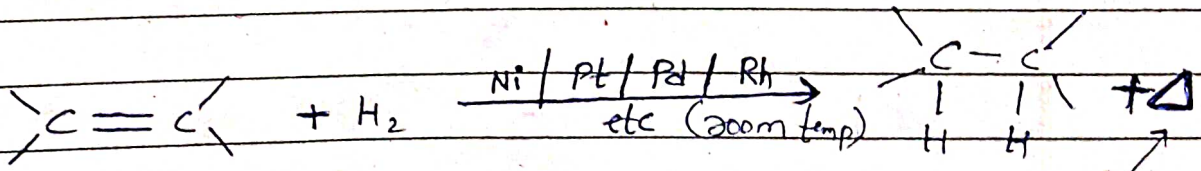
(II) Monochlorination of saturated Carbon.

(III) Ozonolysis of alkenes and alkynes.

(IV) Practical organic chemistry {POC}

- Qualitative analysis
  - Test of functional group
  - Test of elements (N, S, X, P)
- Quantitative analysis

→ Hydrogenation of alkenes & alkynes.
↳ addition of H_2 on C=C, C≡C.



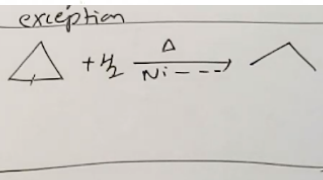
$$-C\equiv C- \;+\; 2H_2 \longrightarrow -\overset{H}{\underset{H}{C}}-\overset{H}{\underset{H}{C}}- \;+\; \Delta$$

* moles of H_2 reacted = no. of C-C π bonds in per molecule of Compound.

* In presence of above metal Catalyst Complete hydrogenation take place on alkyne.

* Hydrogenation of aromatic ring is possible only on high temperature.

exception



Q. Identify all possible structure of reactants alkene for following reactions :-

(I) Hydrogenation of Alkenes and Alkynes

Q Identify all possible structures of reactants (Alkene) for following reactions.

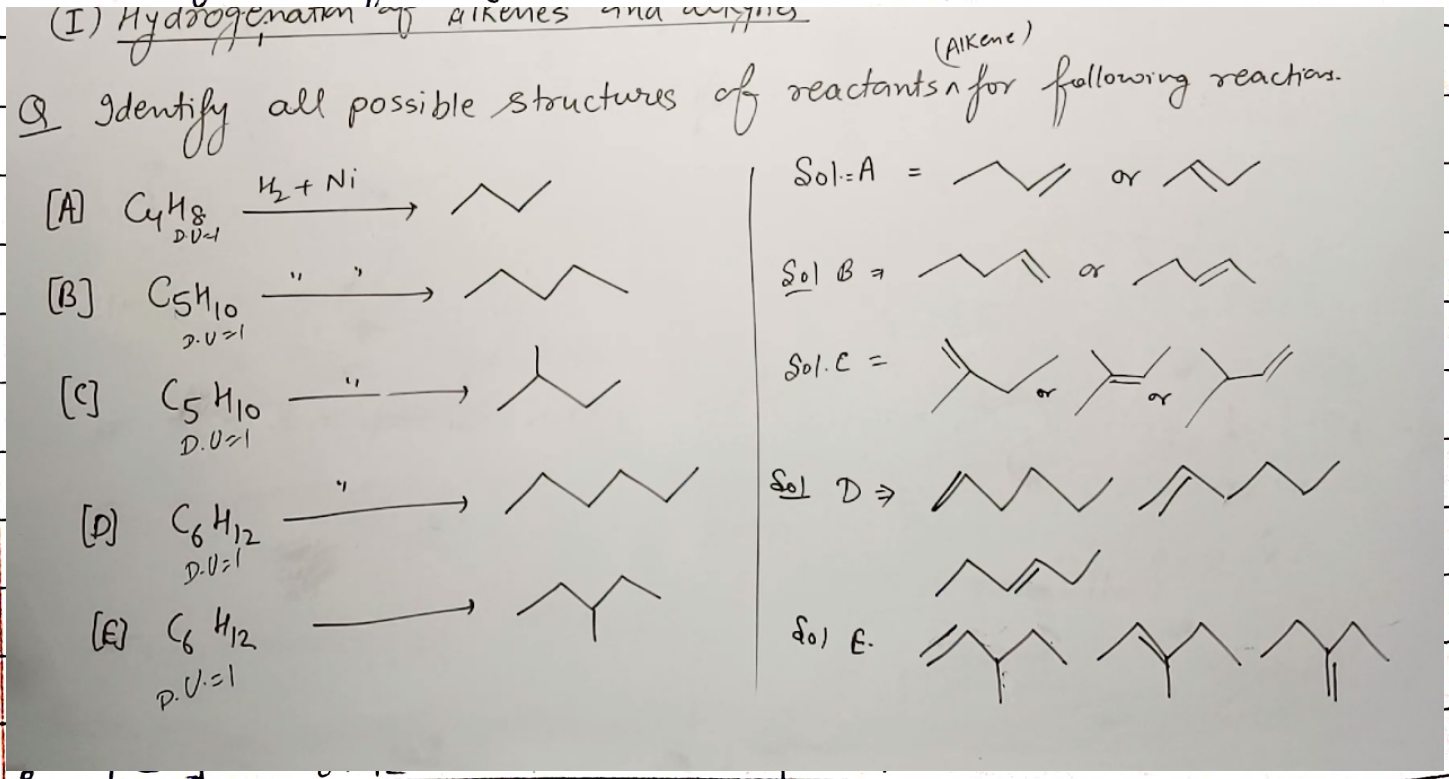
[A] C_4H_8 (D.U=1) $\xrightarrow{H_2 + Ni}$

[B] C_5H_{10} (D.U=1) $\xrightarrow{\quad "\quad}$

[C] C_5H_{10} (D.U=1) $\xrightarrow{\quad "\quad}$

[D] C_6H_{12} (D.U=1) $\xrightarrow{\quad "\quad}$

[E] C_6H_{12} (D.U=1) $\longrightarrow$



Sol:- A = or

Sol B = or

Sol. C = or or

Sol D =>

Sol E.

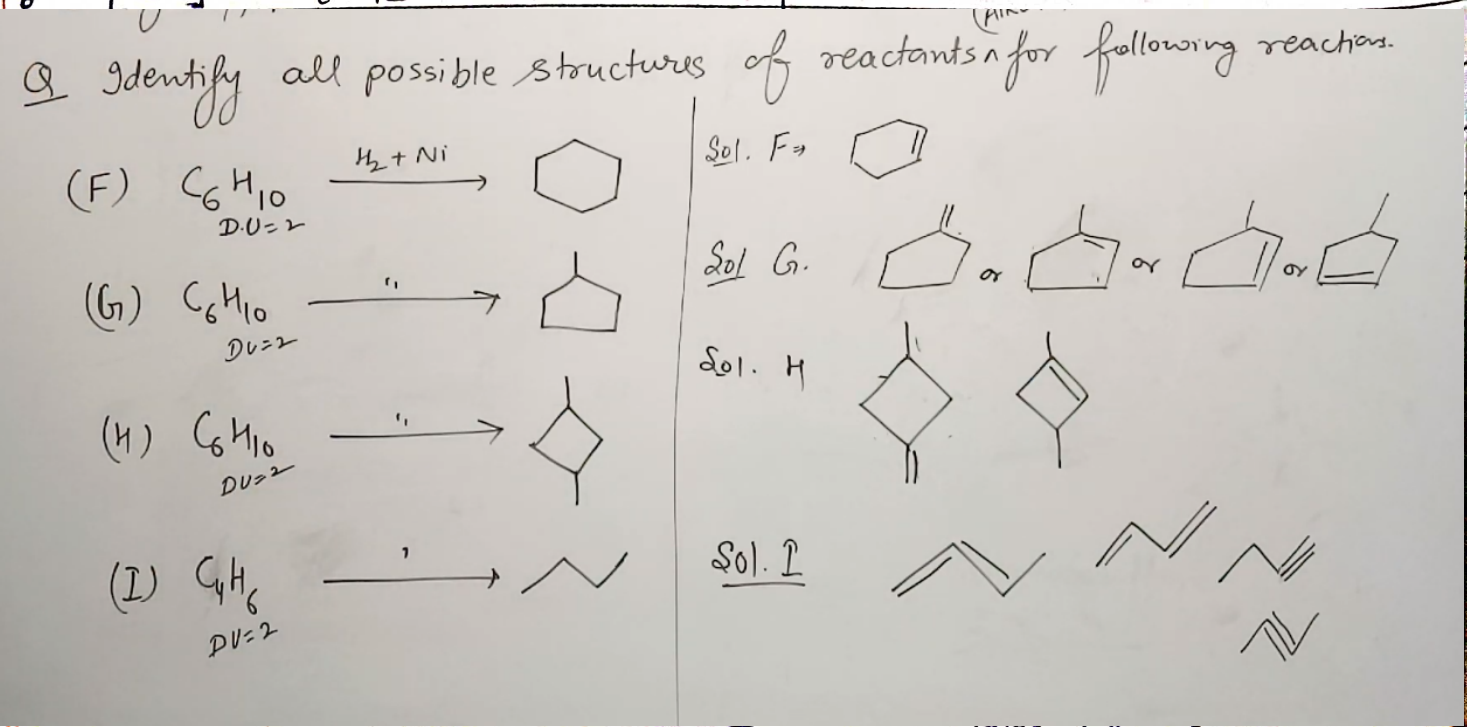
Q Identify all possible structures of reactants (Alkene) for following reactions.

(F) C_6H_{10} (D.U=2) $\xrightarrow{H_2 + Ni}$

(G) C_6H_{10} (D.U=2) $\xrightarrow{\quad "\quad}$

(H) C_6H_{10} (D.U=2) $\xrightarrow{\quad "\quad}$

(I) C_4H_6 (DU=2) $\xrightarrow{\quad "\quad}$



Sol. F =>

Sol G. or or or

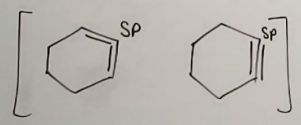
Sol. H

Sol. I

Try it again

(J) C_6H_8 (DU=3) ⟶

Sol



Unstable high ring Strain

SP ⇒ Bond angle = 180°

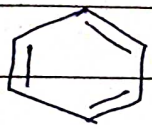
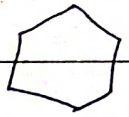
Note ⇒ sp hybridized carbon is unstable in small (upto 7 carbon atom) cyclic compounds (due to ring strain)

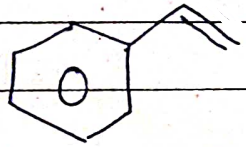
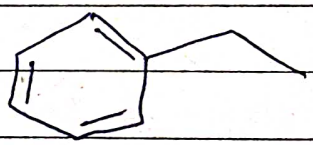
Date :

unstable in Cyclic n}.

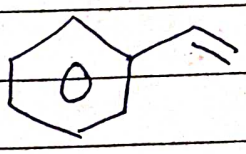
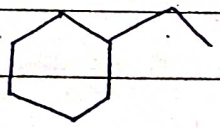
* $\xrightarrow[\text{(Room temp)}]{3H_2 + Ni + \Delta}$ No Reaction

{Aromatic high stable comp.}

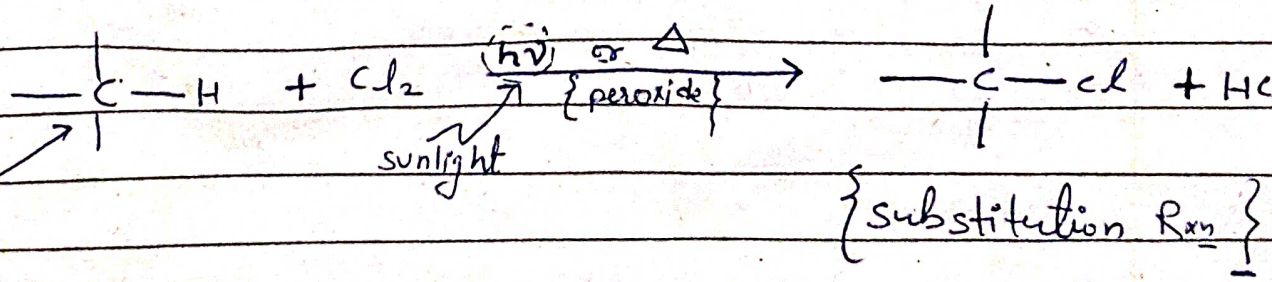
*  $\xrightarrow{3H_2 + Ni + \Delta}$  → cyclohexane

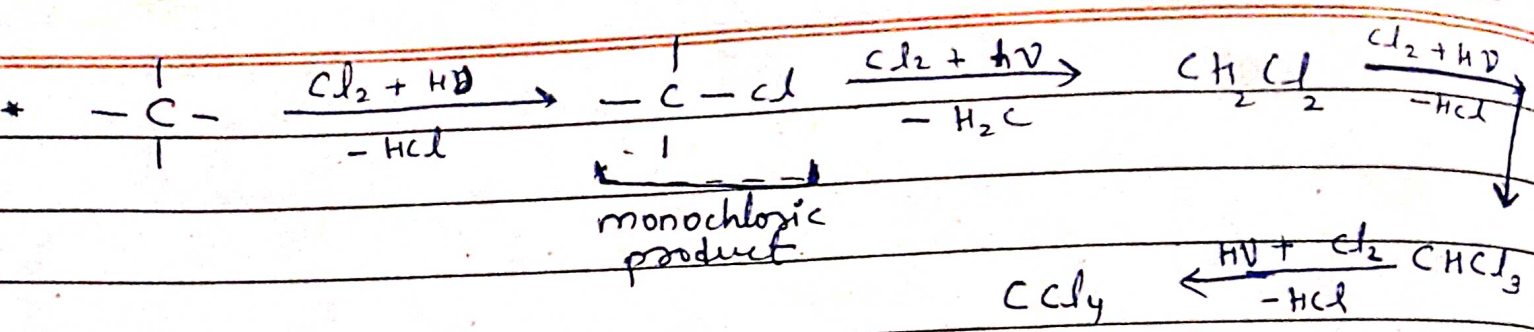
*  $\xrightarrow[\text{Room temp}]{H_2 + Ni}$ 

(Styrene)

*  $\xrightarrow{H_2 + Ni + \Delta}$ 

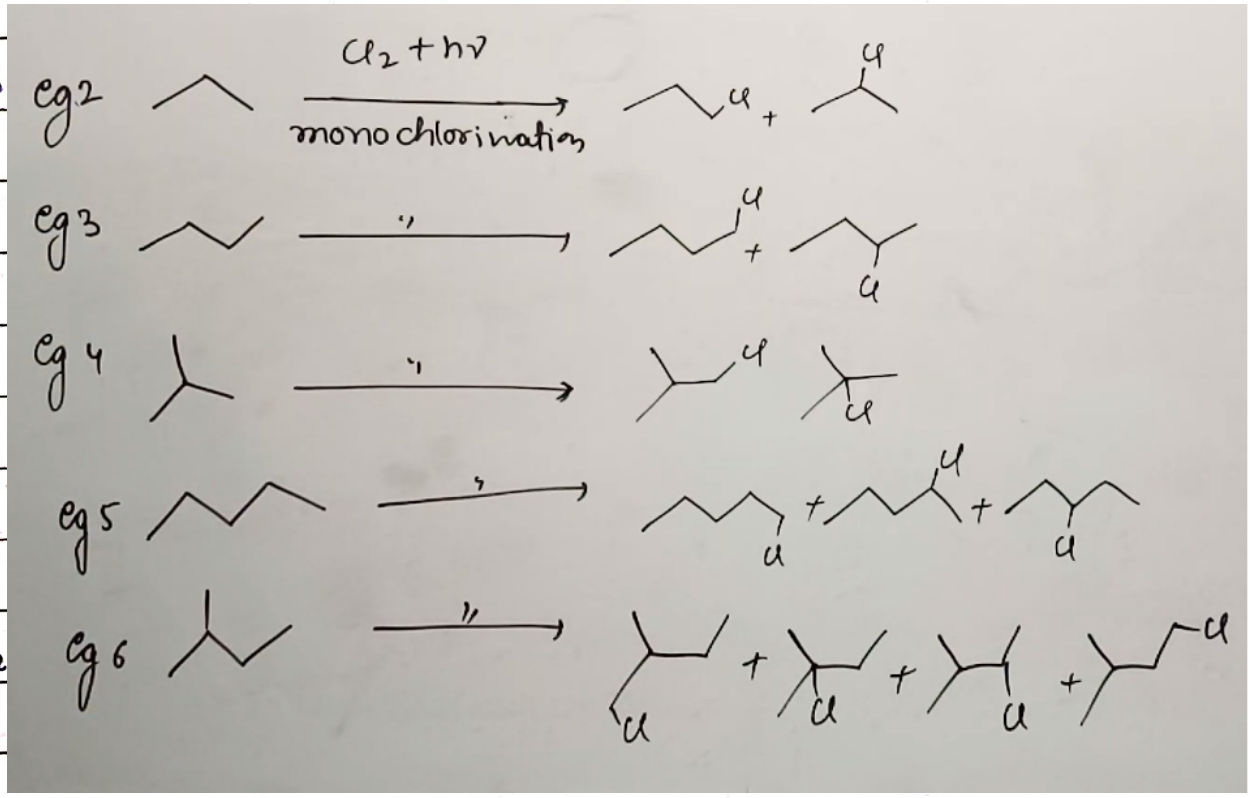
→ Monochlorination of saturated Carbons.

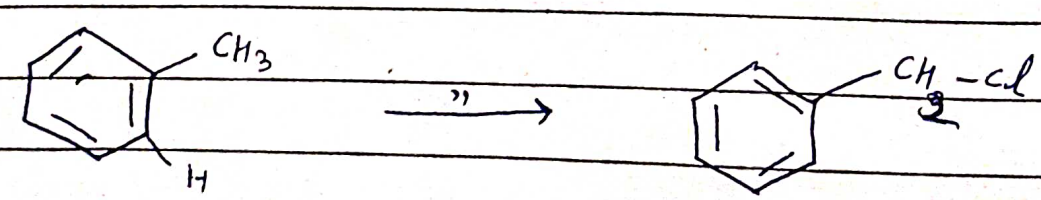


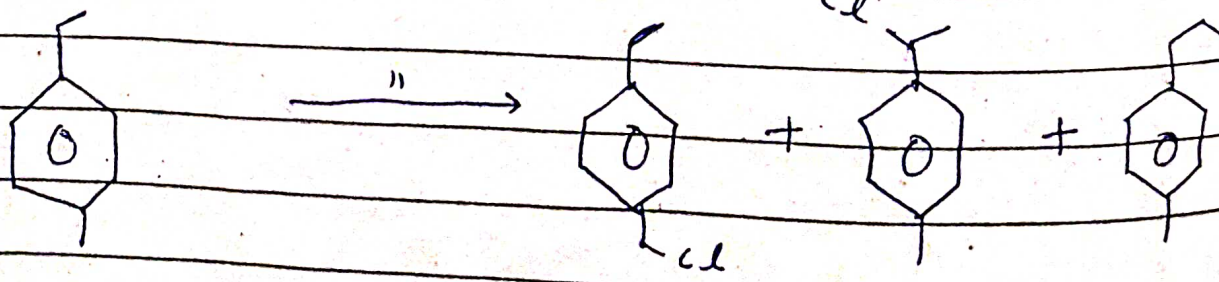


* Possibility for formation of monochloro product in major amount.

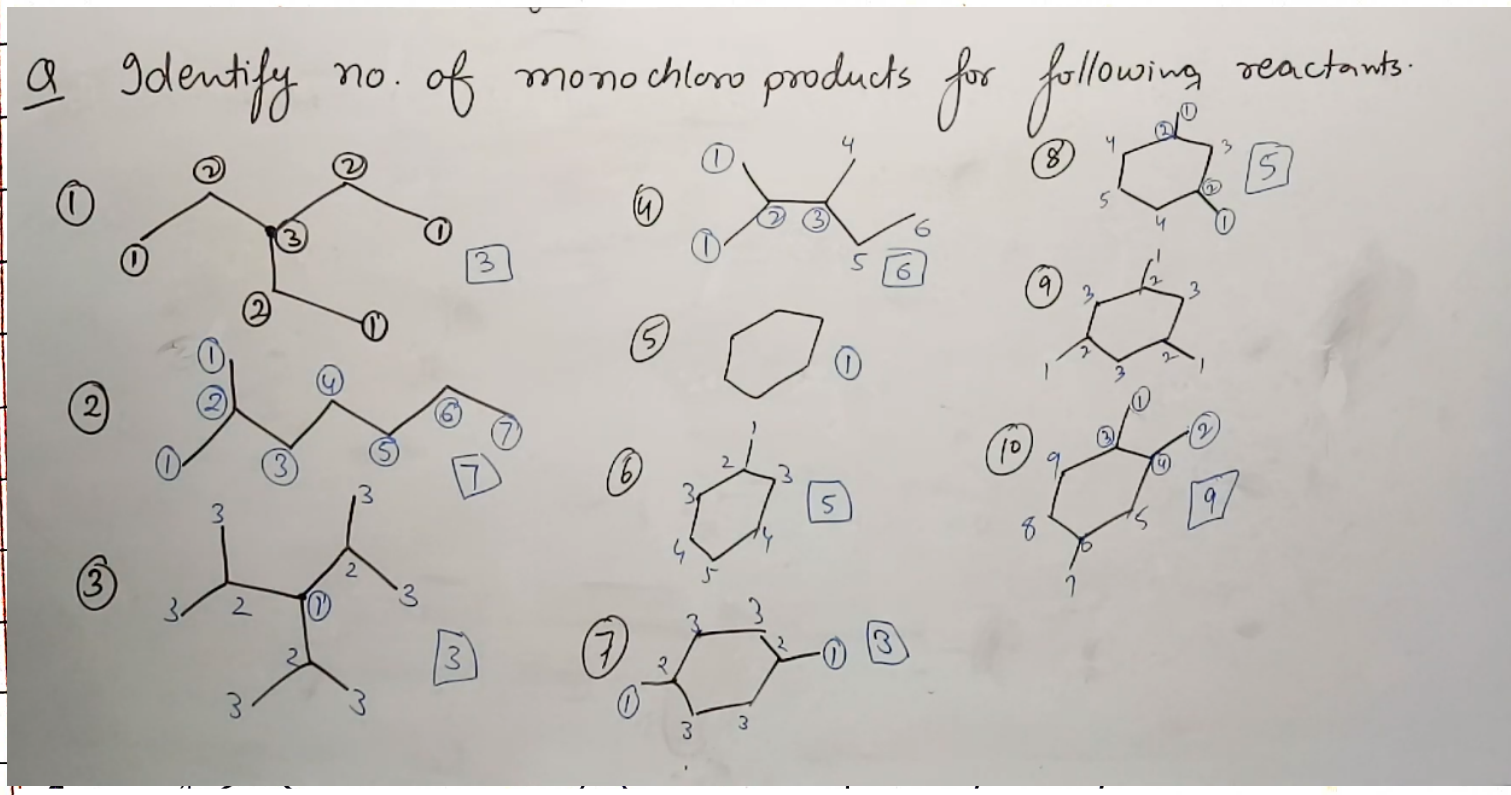
→ amount of alkane > amount of Cl_2

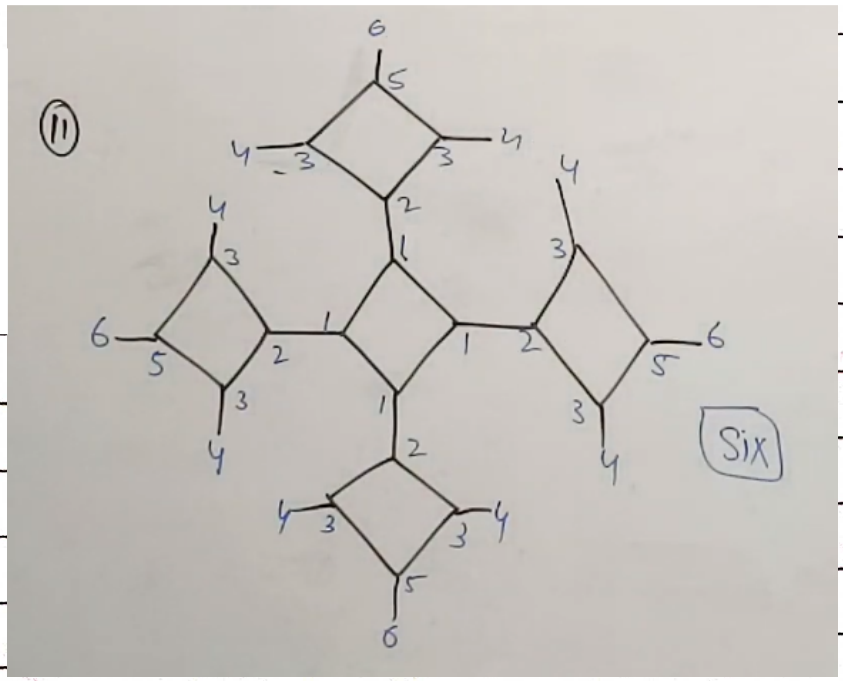






Q. Identify no. of monochloro products for following reactants.





→ Ozonolysis of alkenes & alkynes

↳ Dissociation of C=C or C≡C with reaction of O_3.

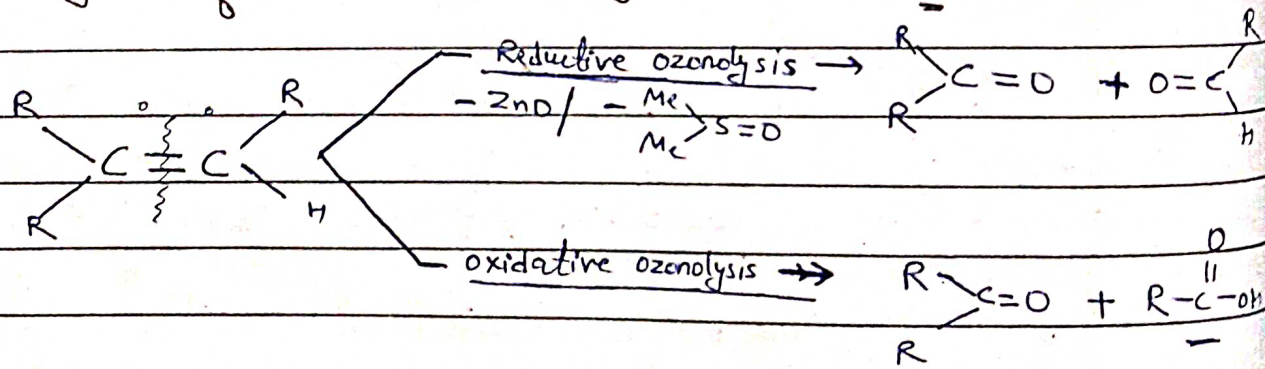
Types of Ozonolysis

- Reductive ozonolysis
- oxidative ozonolysis

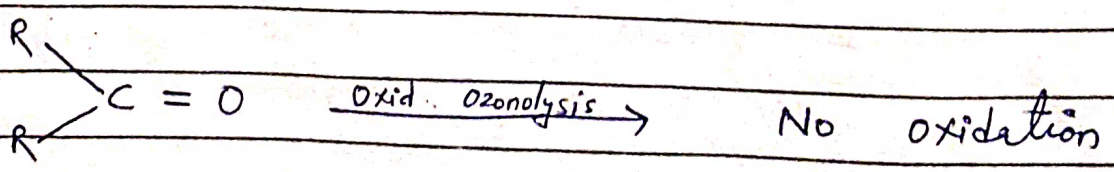
• Reagents
i) O_3 ii) $Zn + H_2O$ / Me_2S

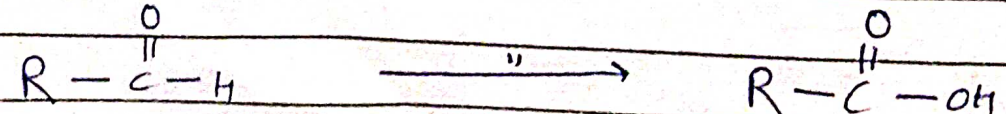
• Reagents
i) O_3 ii) (H_2O / H_2O_2)

* Ozonolysis of alkenes { General rxn }



* Hint for oxidative ozonolysis.





- $R-\underset{\text{adjacent dione}}{\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-R} \xrightarrow{[O]} R-\overset{O}{\overset{\|}{C}}-OH + HO-\overset{O}{\overset{\|}{C}}-R$

- $R-\overset{O}{\overset{\|}{C}}-OH \xrightarrow{[O]}$ No Reaction.

- $H-\overset{O}{\overset{\|}{C}}-OH \xrightarrow{[O]} \left[H-O-\overset{O}{\overset{\|}{C}}-OH\right] \longrightarrow H_2O + CO_2$

{formic acid} — Carbonic acid (unstable)

* Ozonolysis of alkynes {general rxn}

$R-C\equiv C-R'$

→ Red. ozonolysis → $-ZnO \,/\, Me_2S=O$ → $R-\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-R'$ {adjacent dione}

→ Oxi. Ozonolysis → $R-\overset{O}{\overset{\|}{C}}-OH + R'-\overset{O}{\overset{\|}{C}}-OH$

* Ozonolysis of aromatic rings not possible at room temperature. It is possible only on high temp.

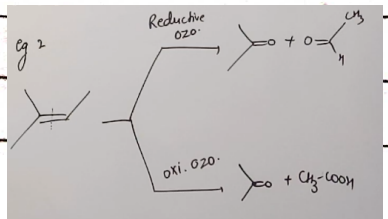
* We can identify position of Carbon - Carbon multiple bonds in reactant on the basis of structure of products.

eg 1.

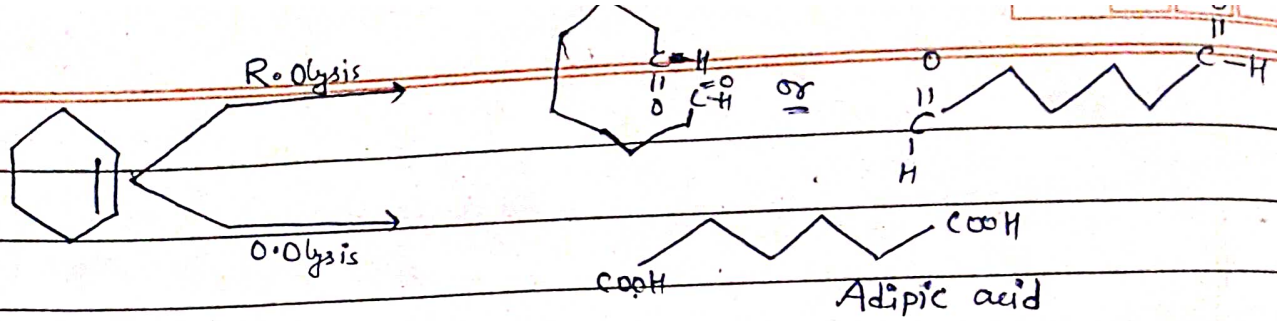
Red. ozonolysis → $=O + O=$

Oxi. Ozonolysis → $=O + O=$

eg 2

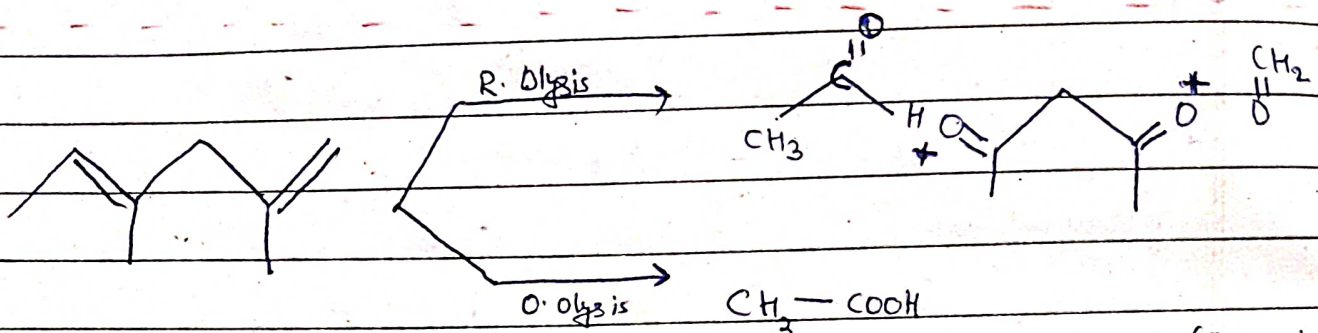


2.



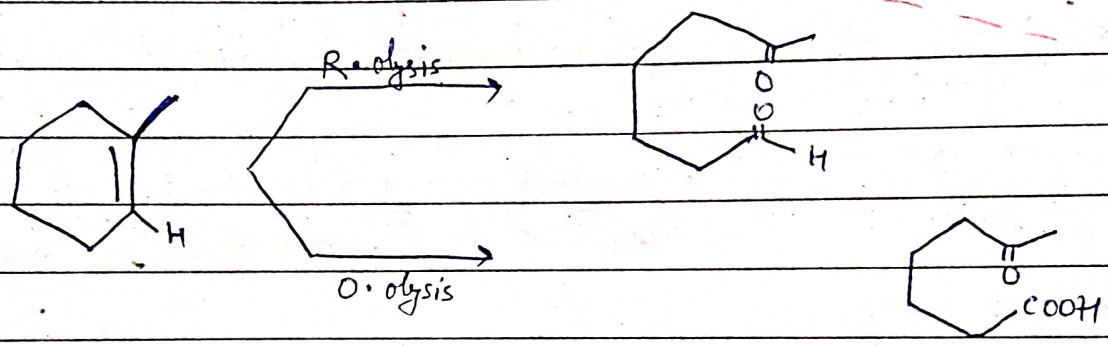
Adipic acid

3.

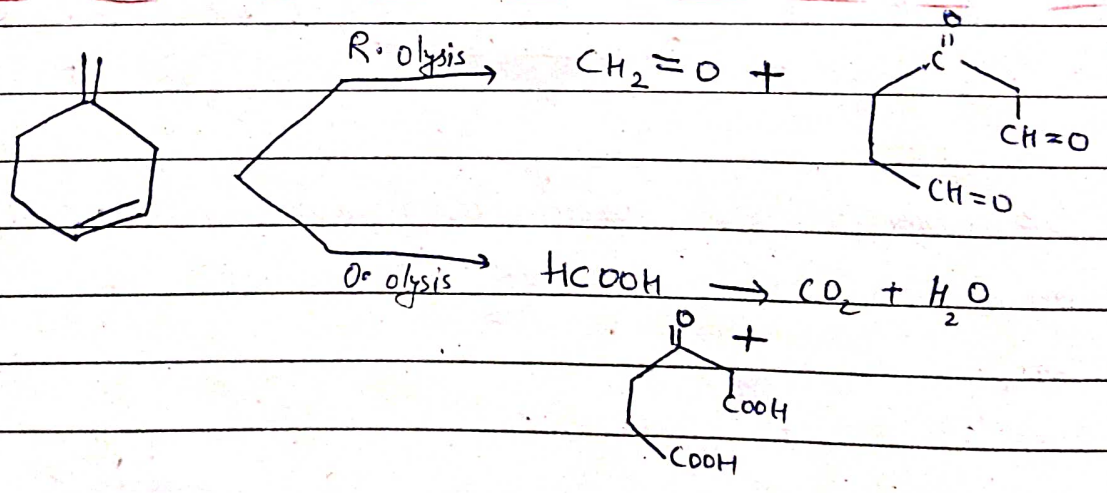


$CH_3 - COOH + H - COOH \rightarrow CO_2 + H_2O$

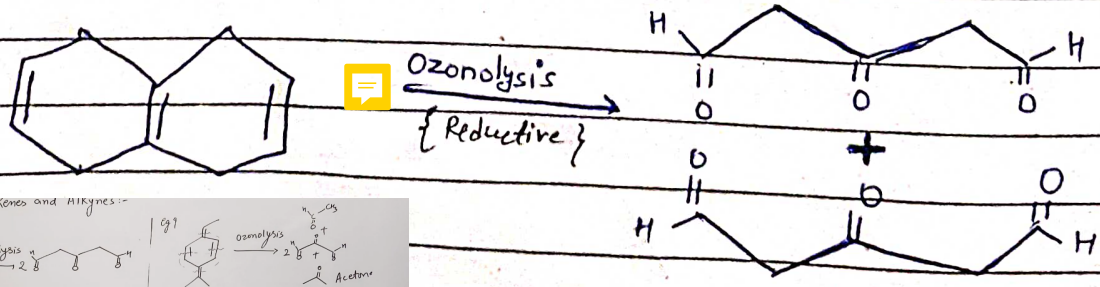
4.

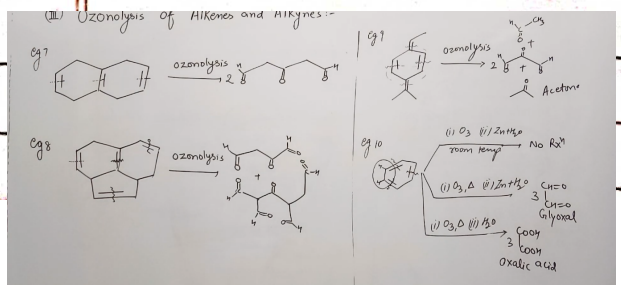


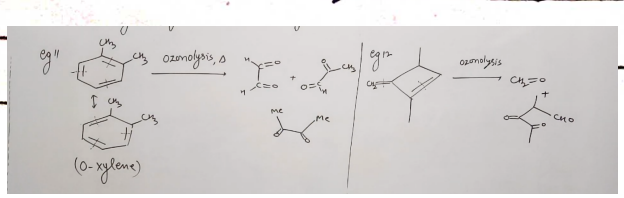
5.



6.



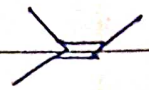


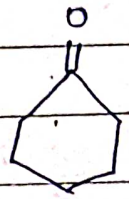
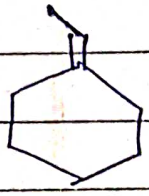


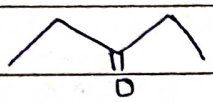
Date :

Q. Identify all possible structure of reactants.

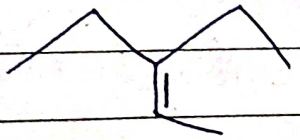
1. [A] $\xrightarrow{\text{Ozonolysis}}$ Acetone. →

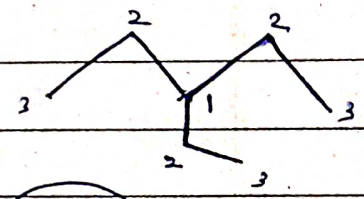
2. [B] $\xrightarrow{''}$ + $CH_3-CH=O$ → 

3. [C] $\xrightarrow{''}$ $CH_3-CH=O$ +  → 

4. [D] $\xrightarrow{''}$  + $CH_3-CH=O$

$H_2 + Ni$ → [E] $\xrightarrow{Cl_2 + h\nu}$ n = no. of monochloro stru. iso

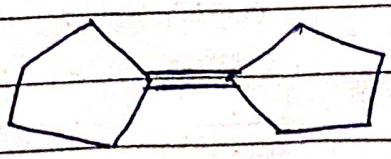
(D) → 

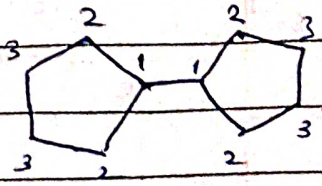
(E) → 

n = 3

5. [F] $\xrightarrow{O_3}$

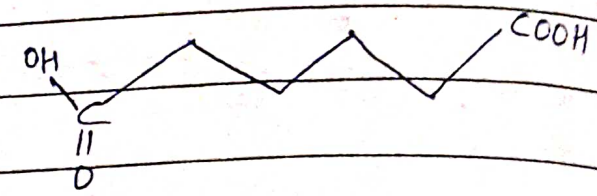
$H_2 + Ni$ → [G] $\xrightarrow{Cl_2 + h\nu}$ m = no. of monochloro structural iso

(F) → 

(G) → 

n = 3

Q. Identify all possible structure of reactant.

1. C_6H_{10} [A] —Ozonolysis→ 

oxidative ozonolysis

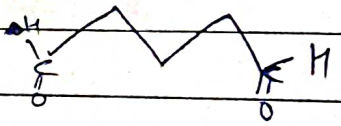
Ans A →

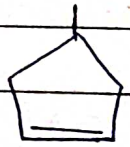
2. C_4H_8 [B] —"→ $2CH_3-COOH$
oxidative ozonolysis

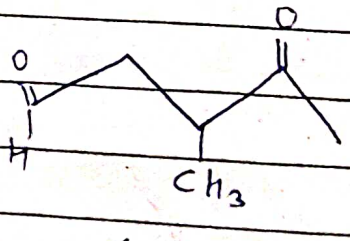
Ans. B →

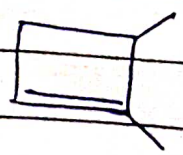
3. C_4H_6 [C] —"→ $2CH_3-COOH$
oxidative ozonolysis

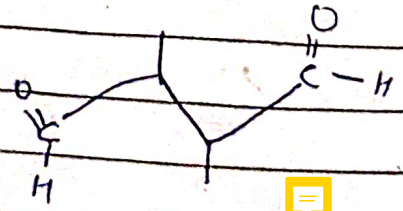
Ans C → $CH_3-C\equiv C-CH_3$

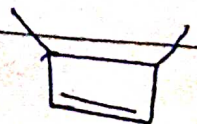
4. C_6H_{10} [D] —"→ 

Ans D → 

5. C_6H_{10} [E] —"→ 

Ans E → 

6. C_6H_{10} [F] —"→ 

Ans [F.] → 

7. C_6H_{10} [G] $\longrightarrow$ $2CH_3-CH=O + CH(=O)-CH=O$

Ans G $\rightarrow$ $CH=CH-CH_3$ | $CH=CH-CH_3$

8. C_5H_8 [H] $\xrightarrow{''}$ $CH_3-\overset{O}{\overset{\|}{C}}H + CH_2=O + CHO-CHO$

Ans H $\rightarrow$ $CH=CH_2$ | $CH=CH-CH_3$

→ Test of functional group.

① Functional group → $>C=C<$ & $-C\equiv C-$ Alkene & alkyne {Non-aromatic}

Test & Reagents →

a: Baeyer's Reagent test.

→ Cold dil $KMnO_4 \cdot \overset{\ominus}{O}H$ {Purple}

$>C=C< \xrightarrow[\text{Cold}]{KMnO_4 \cdot OH} >C(OH)-C(OH)<$

$-C\equiv C- \xrightarrow{''} -\overset{O}{\overset{\|}{C}}-\overset{O}{\overset{\|}{C}}-$

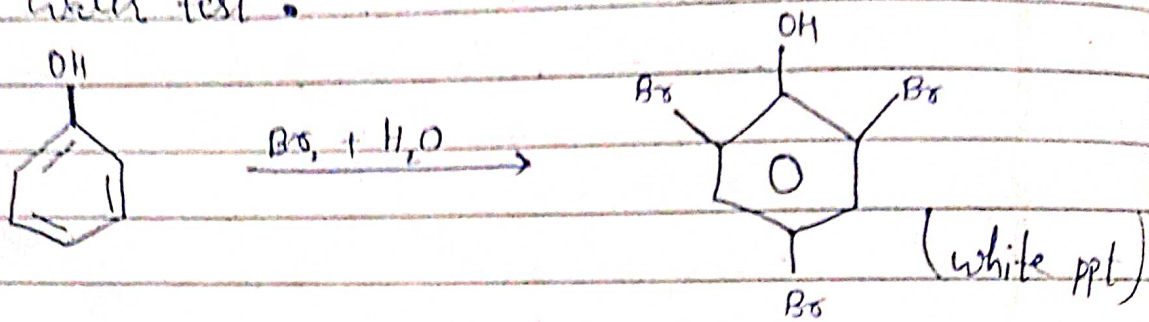
observation → Colours Dissapears.

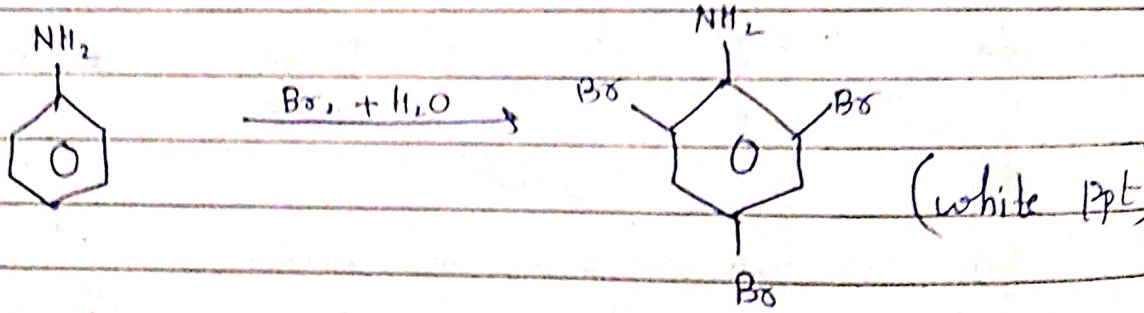
b. Bromine water test
↳ ($Br_2 + H_2O$) Reddish - Brown

$$>C=C< \xrightarrow{Br_2 + H_2O} >C(Br)-C(OH)<$$

Observation → Colour Disappears.

Note:- Phenol & Aniline are also give bromine water test.





2. → $-C\equiv C^{\delta-} \leftarrow H^{\delta+}$ {Terminal alkyne}.

a. Tollen's Regent test

$$AgNO_3 + NH_4OH \longrightarrow [Ag(NH_3)_2]OH$$

{Ammonical silver nitrate}

$$R-C\equiv C-H \longrightarrow R-C\equiv C^{-}Ag^{+}\downarrow$$

white ppt

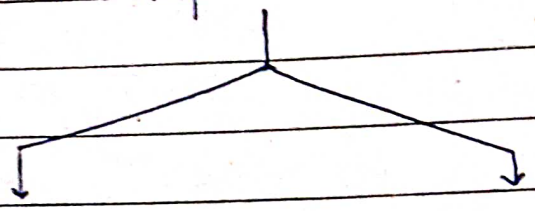
b. $CuCl_2 + NH_4OH$:-

$$R-C\equiv C-H \xrightarrow{CuCl_2 + NH_4OH} R-C\equiv C^{-}Cu^{+}\downarrow$$ Red P

3. → $R-\overset{O}{\overset{\|}{C}}-OR'$ {esters}

a. Smell test → fruity smell

b. Phenolphthalein test



in Basic {Pink}

in Acidic {Colourless}

c. Hydrolysis

$$R-\overset{O}{\overset{\|}{C}}-O-R' + H_2O \xrightarrow[\text{HPH}\ \Delta]{\text{dil NaOH +}} R-\overset{O}{\overset{\|}{C}}-OH + R'-OH$$

Pink

Colourless solution after some time.

4. → $-COOH$ {Carboxylic acid}

a. Litmus test {Blue} → Blue to red

b. $NaHCO_3$ solution test {Colourless}

$$R-COOH + NaHCO_3 \longrightarrow R-COONa + H_2CO_3$$

$$H_2CO_3 \downarrow \; CO_2\uparrow + H_2O$$

Colourless + odourless gas gives lime water test

Date :

c. Esterification

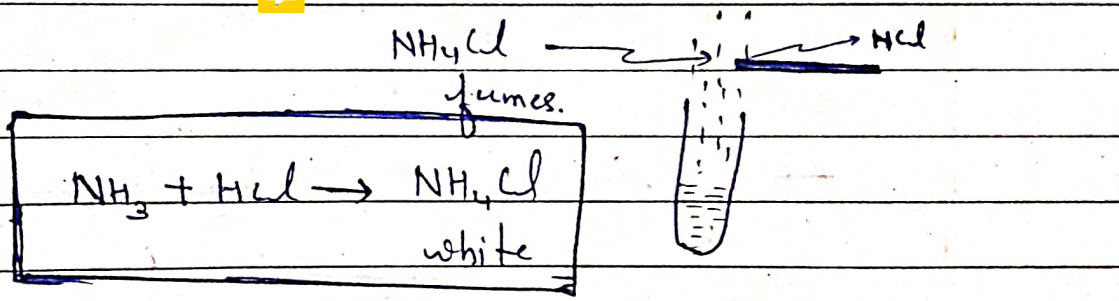
$$R-COOH + HO-Et \xrightarrow[-H_2O]{\text{Conc. } H_2SO_4,\ \Delta} R-\overset{O}{\overset{\|}{C}}-O-Et$$

fruity smell

5. Amide $R-\overset{O}{\overset{\|}{C}}-NH_2$

a. Hydrolysis Test

$$R-\overset{O}{\overset{\|}{C}}-NH_2 + H_2O \xrightarrow[\Delta]{\text{dil NaOH}} R-\overset{O}{\overset{\|}{C}}-ONa + NH_3\uparrow$$



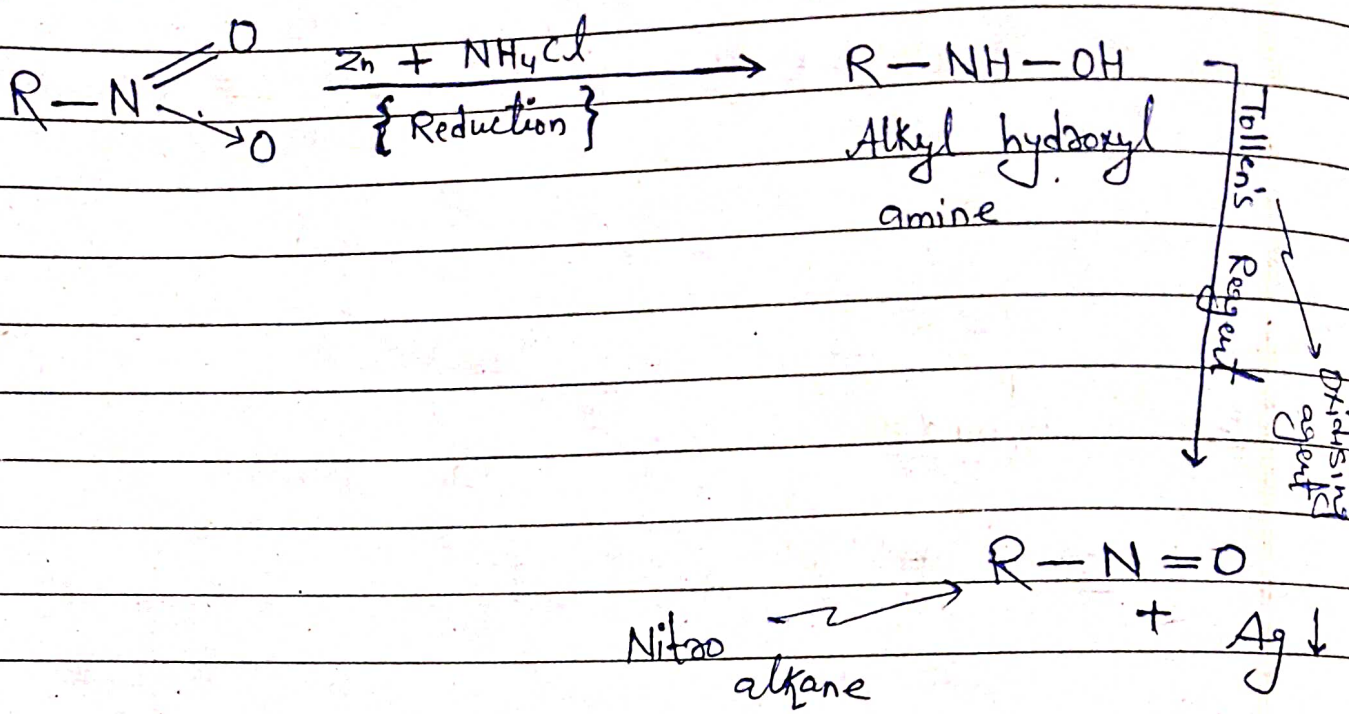
observation → NH_3 (smell) { give white fumes with HCl dipped glass rod }.

6. Nitro $-N\overset{O}{\underset{O}{=}}$

a. Mulliken's Barker test.

i) $Zn + NH_4Cl$ ii) Tollen's Reagent.

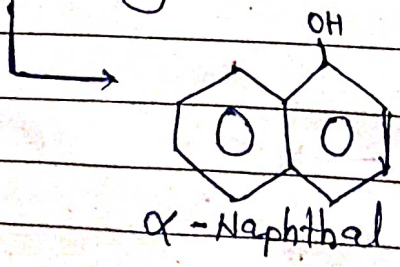
{ $AgNO_3 + NH_4OH$ }.



Observation :- Silver mirror in glass tube walls.

7. → Carbohydrate ~~Poly hydrate~~ { Polyhydroxy Carbonyl Comp. }

a. Molish Reagent test :-



+ C_2H_5OH → Violet colour

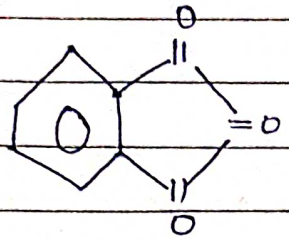
b. Barfoed test :- { generally given by monosaccharide }

$(CH_3COO)_2Cu$ + Sugar $(R-CHO)$ → $R-COOH + Cu_2O$

Brick red

8. Amino Acid { α- amino Carboxylic acid }.

a. Ninhydrin



Violet / Purple.

b. Xanthoproteic test { Test of proteins / Test of aromatic amino acids.

Conc. HNO_3 · Δ ⟶ NaOH Yellow → Orange

c. Biuret test

9. Phenolic −OH { Ar−OH }.

a. Blue litmus test Blue → red.

b. Bromine Water test white ppt

c. Neutral $FeCl_3$ Test :-

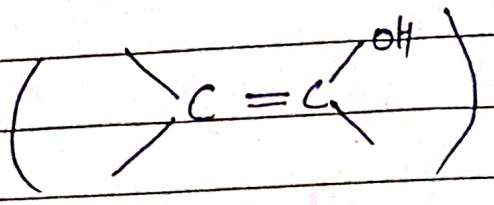
→ Brown Soln.

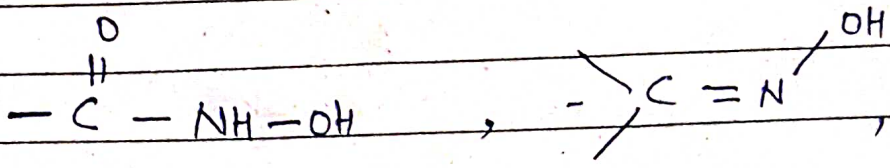
$$FeCl_3 + Ph\text{-}OH \longrightarrow [Fe(OPh)_6]^{-3}$$

Violet Complex.

Obs observation · Violet / Green / Blue Complex.

Note :- $FeCl_3$ test is also given by

enol 



Hydroxymic acid

Oxime

d. Cerric ammonium nitrate.
{CAN}

$(NH_4)_2[Ce(NO_3)_6]$

Yellow.

Brown - black soln

10. Alcohols.

a. CAN test :- 1°, 2°, 3° alcohols.

Red - pink soln.

b. Lucas reagent test.

→ anhydrous $ZnCl_2$ + Conc. HCl

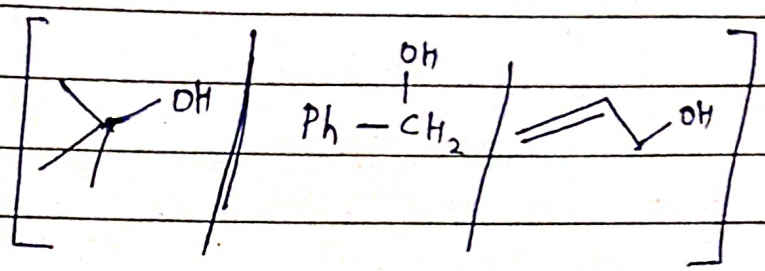
{Useful for distinguish of 1° 2° & 3° alcohols}

$R-OH \xrightarrow[(SN)]{HCl + ZnCl_2} R-Cl + H_2O$

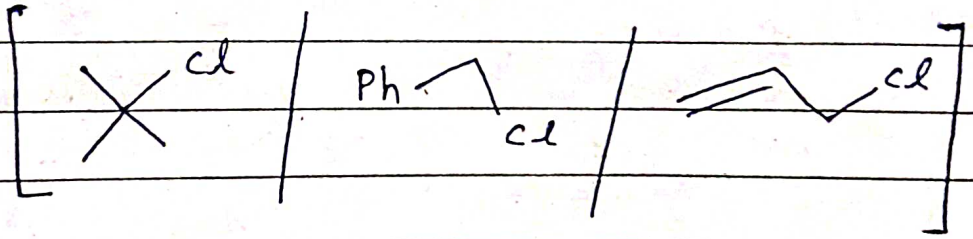
{white turbidity}

$3^\circ > 2^\circ > 1^\circ$ → Reactivity

3° alcohol
Benzylic allylic ale

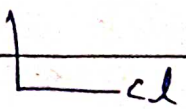


L.R



Instant white Turbidity

2° alcohol —LR→ white turbidity after 5–10 min

1° alcohol —LR→  white turbidity after 30 min or on heating.

C. Victor - meyer test {useful for distinguish of 1°, 2°, 3° alcohols}

1° → $R-\underset{\underset{OH}{|}}{CH_3} \xrightarrow{P+I_2} R-\underset{\underset{I}{|}}{CH_2} \xrightarrow[-AgI\downarrow]{AgNO_2} R-CH_2-NO_2 \xrightarrow[-H_2O]{O=N-OH} R-\overset{\overset{N-OH}{\|}}{C}-NO_2 \xrightarrow{NaOH} R-\overset{\overset{N-ONa}{\|}}{C}-NO_2$

Red.

2° → $\begin{matrix} R \\ R \end{matrix}\!\!>\!\!OH \xrightarrow{P+I_2} \begin{matrix} R \\ R \end{matrix}\!\!>\!\!CH-I \xrightarrow[-AgI]{AgNO_2} \begin{matrix} R \\ R \end{matrix}\!\!>\!\!CH-NO_2 \xrightarrow[-H_2O]{HO-N=O} \begin{matrix} R \\ R \end{matrix}\!\!>\!\!\overset{\overset{N=O}{|}}{C}-NO_2 \xrightarrow{NaOH}$ No Rxn

Blue

3° → $\begin{matrix} R \\ R \\ R \end{matrix}\!\!>\!\!C-OH \xrightarrow{P+I_2} \begin{matrix} R \\ R \\ R \end{matrix}\!\!>\!\!C-I \xrightarrow[-AgI]{AgNO_2} R-\underset{\underset{R}{|}}{\overset{\overset{R}{|}}{C}}-NO_2 \xrightarrow{HNO_2}$ No Rxn $\xrightarrow{NaOH}$ No Rxn

Colourless

RBC Test

Date :

11. → Amine (Test)

a. Isocyanide Test / Carbylamine test } Ali 1° amine / Aro 1° amine

$$R-NH_2 \xrightarrow{CHCl_3 + OH^-} R-NC$$

$$Ar-NH_2 \xrightarrow{CHCl_3 + OH^-} Ar-NC$$

} Isocyanide (foul smell)

b. Hinsberg Reagent ($Ph-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-Cl$, Benzensuphonyl chloride)

{ Test for 1°, 2° amine (Ali or ~~Aro~~ Aro both) }

1° $R-NH_2 + Cl-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-Ph \xrightarrow[-HCl]{} R-\overset{H}{\underset{|}{N}}-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-Ph$

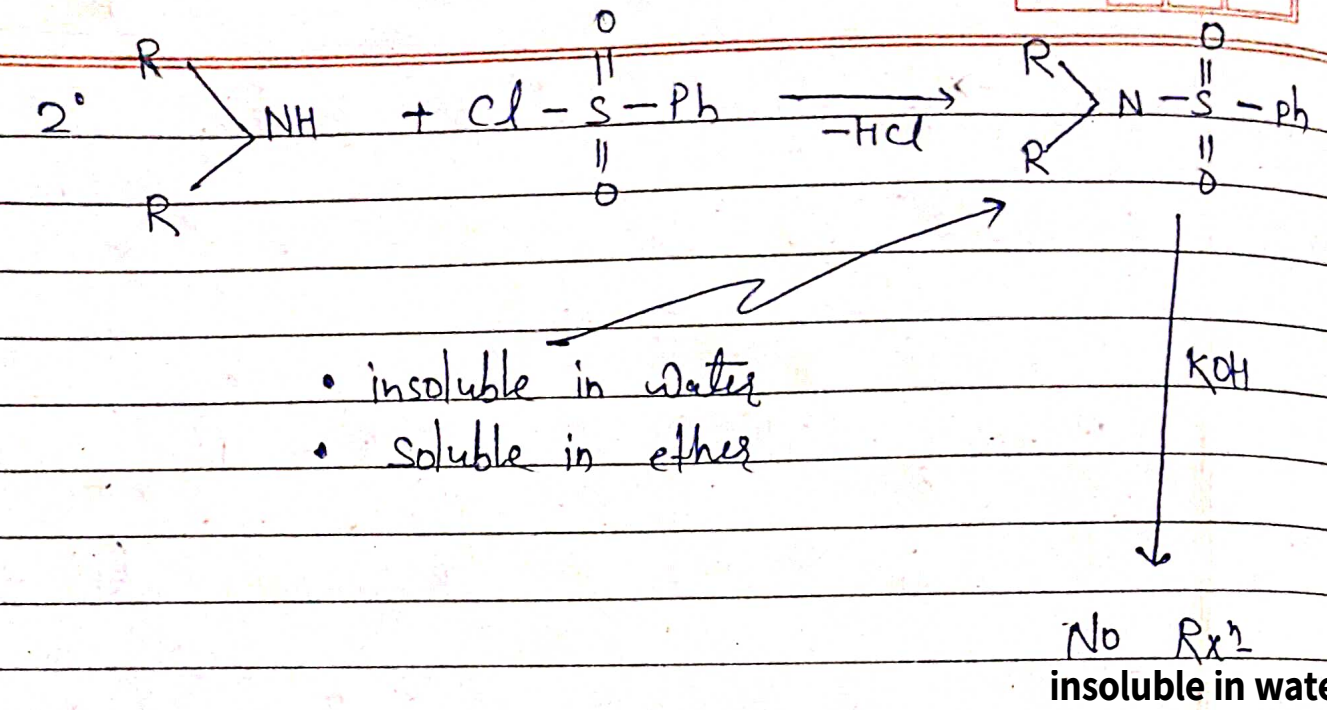
{ amide of sulphonic acid }

- insoluble in water
- soluble in ether

$\xrightarrow[-H_2O]{KOH}$

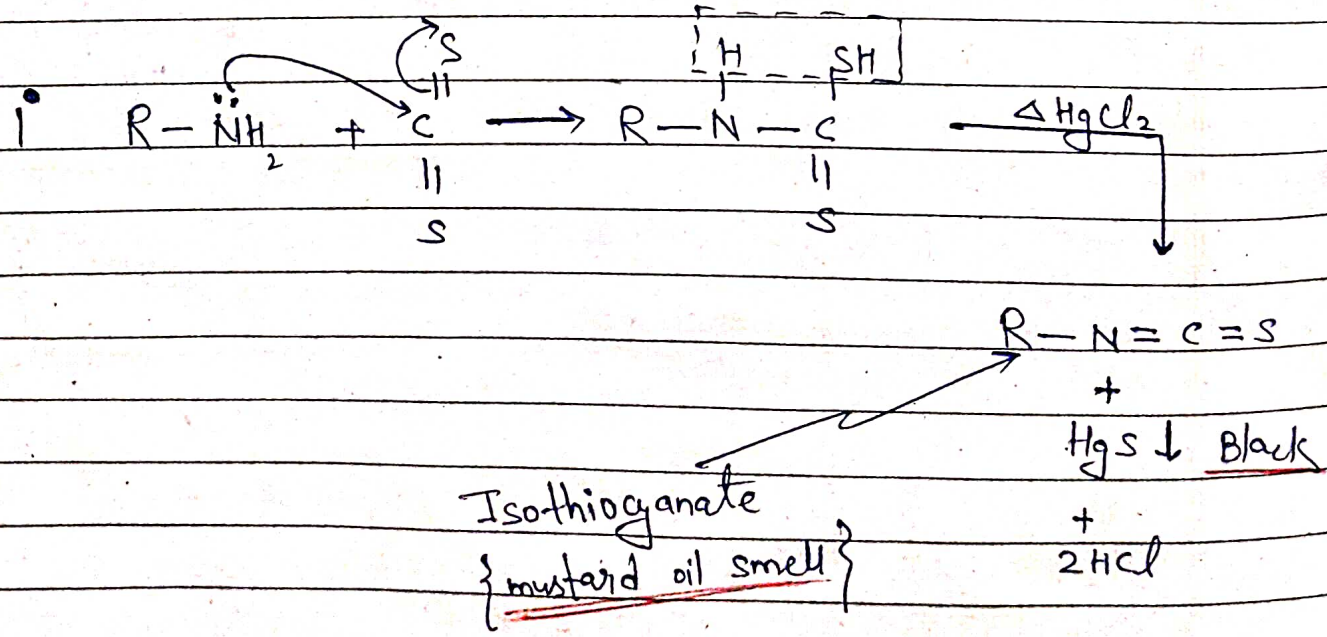
$$R-\overset{-K^+}{N}-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-Ph$$

Soluble in water



3° R_3N + $Cl-SO_2-Ph$ ⟶ No Rx

c. Hafmann's mustard oil test { Test of i (Ali/Ar amine



2° $R_2NH + CS_2$ ⟶ R N – C (SH, S) $\xrightarrow{HgCl_2,\ \Delta}$ No Rx

3° $R_3N + CS_2 \longrightarrow$ No Rxn.

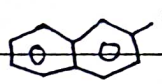
d. Test of 1° aromatic amines {Azodye test}

$$Ar-NH_2 \xrightarrow[0^\circ-5^\circ c]{NaNO_2 + HCl \;(HNO_2)} [Ar-\overset{+}{N}\equiv N]\,Cl^-$$

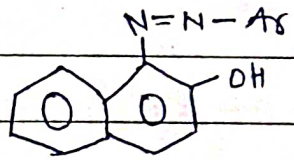
Diazonium salt

$\updownarrow$

$[Ar-N=\overset{+}{N}]\,Cl^-$

$\downarrow$ + 

β- Napthol.

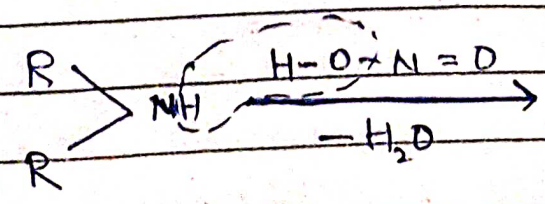


orange-Red dye.

e. Nitrous acid test {for Aliphatic amine}

$$R-NH_2 \xrightarrow[(KNO_2 + HCl)]{HNO_2\,(aq)} R-OH + N_2\uparrow$$

Ali 1° amine

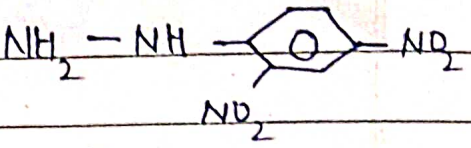
2° amine 

Yellow oily liq.

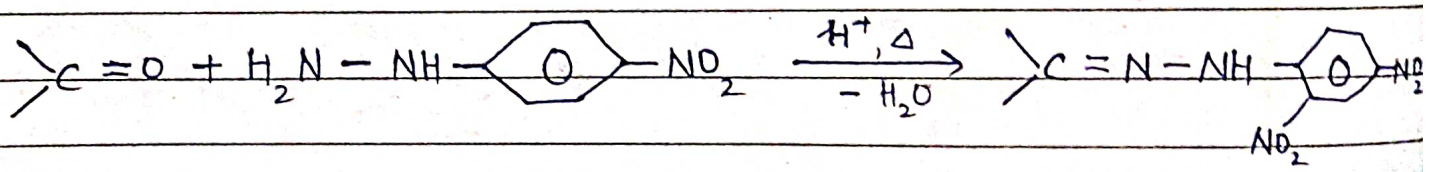
2° $ArNH \xrightarrow{"} Ar_2N-N=O$ Yellow oily liq.

12. → Test of carbonyl Compounds {Aldehyde & ketone}

a. 2,4-DNPH → 2,4-Dinitrophenyl hydrazine =

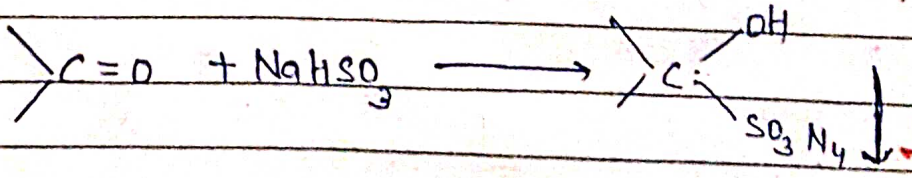


Brady's Reagent



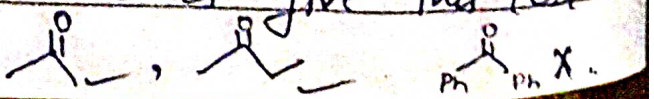
Yellow-orange ppt

b. $NaHSO_3$ {Sod. bisulphate}



→ All aldehydes give this test

→ Small ketones give this test



13. Test of only Aldehydes.

a. Tollen's Reagent test

$R-\overset{O}{\overset{\|}{C}}-H \xrightarrow{oxi} R\text{-}COOH$

$AgNO_3 + NH_4OH \longrightarrow [Ag(NH_3)_2]^+OH^-$

→ mild oxidising agent

$R-\overset{O}{\overset{\|}{C}}-H \xrightarrow{\text{Tollen's reagent}} R-\overset{O}{\overset{\|}{C}}-ONH_4 + Ag\downarrow$

or

$R-COOH$

Silver ~~mirror~~

b. Fehling Solution test

Fehling soln [A] — $(CuSO_4 \cdot NaOH)$ soln → Deep blue

Fehling soln [B] — Sod. pot. Tartrate (Rachelle's salt)

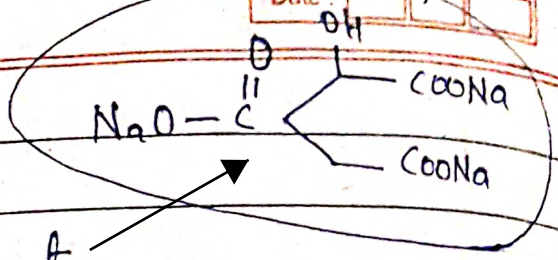
alkaline soln. of CuSO4

Fehling soln [A + B]

$CuSO_4 \xrightarrow{OH} Cu(OH)_2 \longrightarrow CuO$

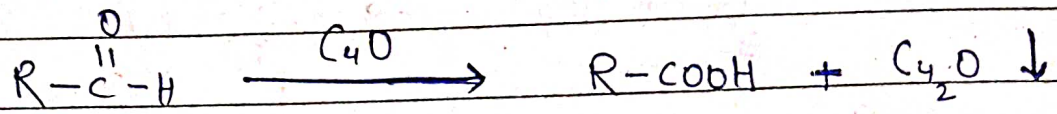
$R-CHO \xrightarrow{+2\ CuO} R-\overset{O}{\overset{\|}{C}}-OH \text{ or } R\text{-}COONa + Cu_2O\downarrow$ Red

C. Benedict Solution

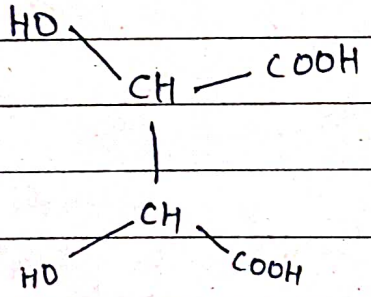


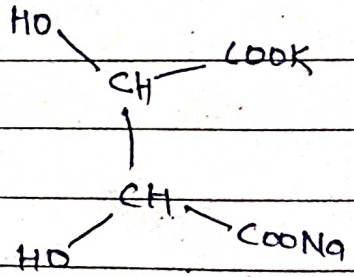
$CuSO_4 + Na_2CO_3 +$ Sod. Citrate

Reaction same as fehling Sol.



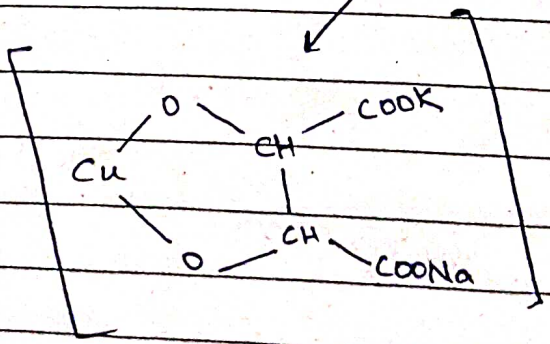
Tartaric acid



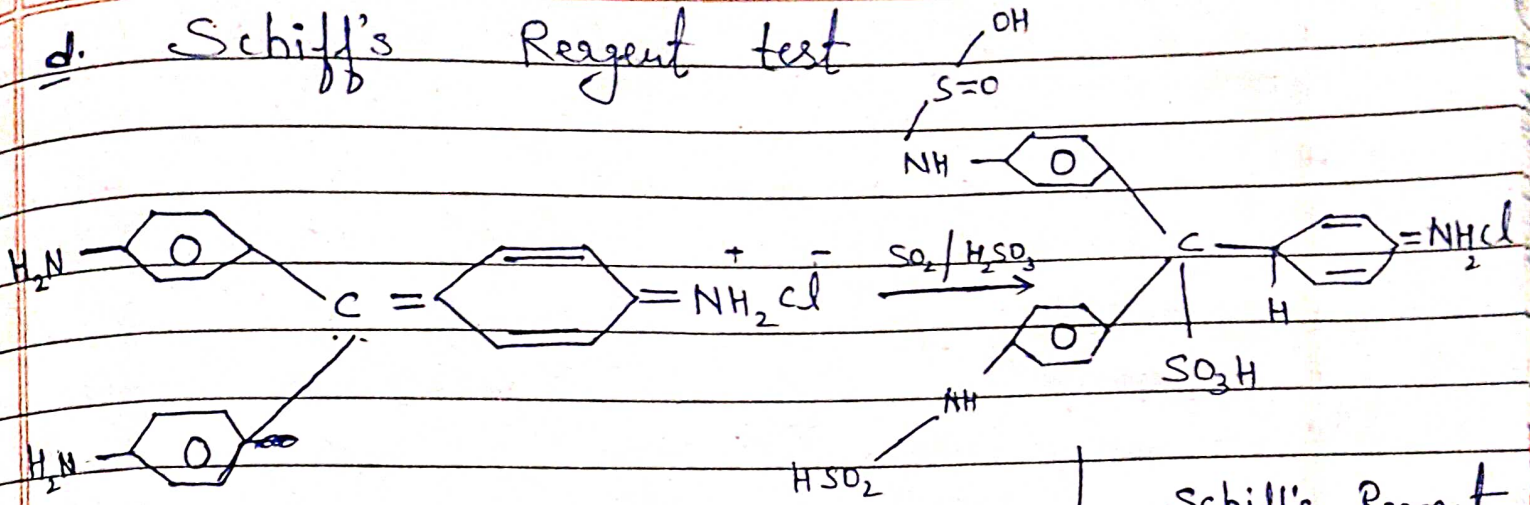


Sol. Pot. Tartrate
{Rachelle salt}

Cu^{+2}



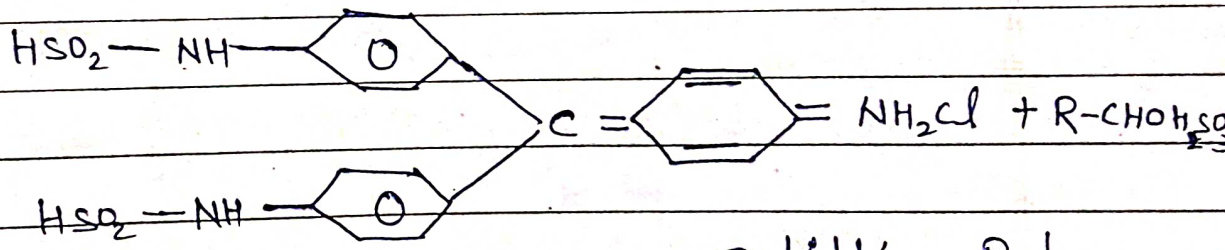
d. Schiff's Reagent test



Rosaniline hydrogen chloride
(Red)

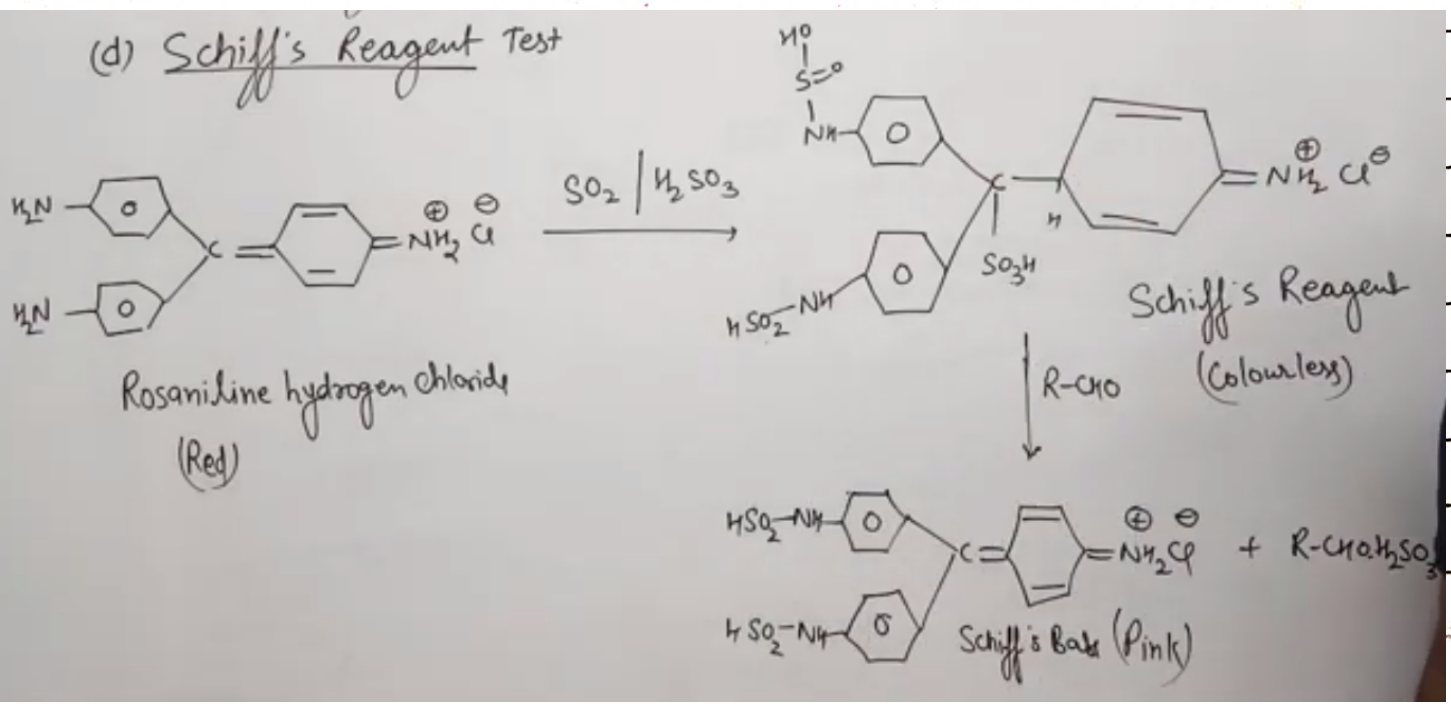
Schiff's Reagent
(Colourless)

R-CHO



Schiff's Base
(Pink)

(d) Schiff's Reagent Test

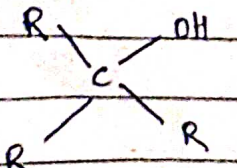


Rosaniline hydrogen chloride
(Red)

Schiff's Reagent
(Colourless)

Schiff's Base (Pink)

Date

Note:		Tollen's test	Fehling or Benedict	Schiff's Reagent
①	$R-CHO$ (Aliphatic)	✓	✓	✓
②	$Ar-CHO$ (Aromatic)	✓	X	X
③	$R-\overset{O}{\overset{\|\|}{C}}-R$ (ketone)	X	X	X
④	$H-\overset{O}{\overset{\|\|}{C}}-OH$ (formic acid)	✓	✓	X
⑤	$\begin{matrix}CHO\\\|\\CHO\end{matrix}$ (glyoxal)	✓	X	✓
⑥	$\begin{matrix}CHO\\\|\\COOH\end{matrix}$ (glyoxalic acid)	✓	X	✓
⑦	$R-NH-OH$ (Alkyl hydroxyl amine)	✓	X	X
⑧	$\begin{matrix}CH_2-OH\\\|\\C=O\\\|\\R\end{matrix}$ (α Hydroxy ketone)	✓	✓	X
⑨	 (Hemiacetal)	✓	✓	X
⑩	$R-C\equiv C-H$	✓	X	X

Note :-

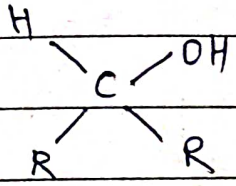
* $H-COOH$

CH_2-OH

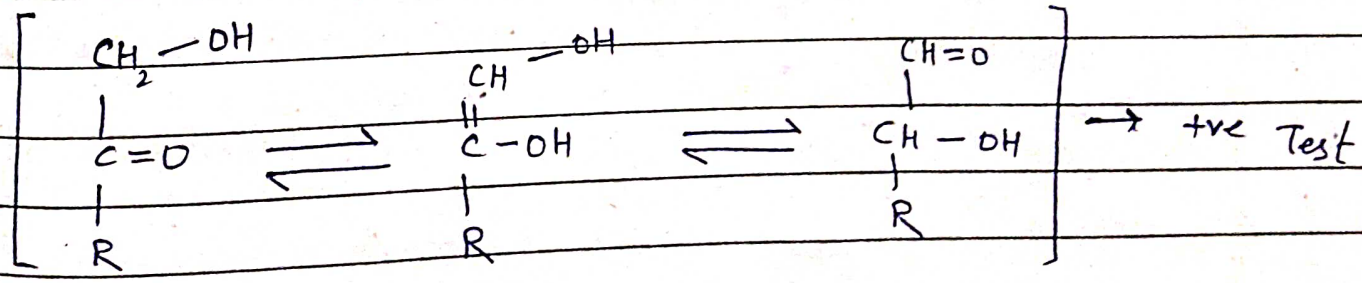
$C=O$

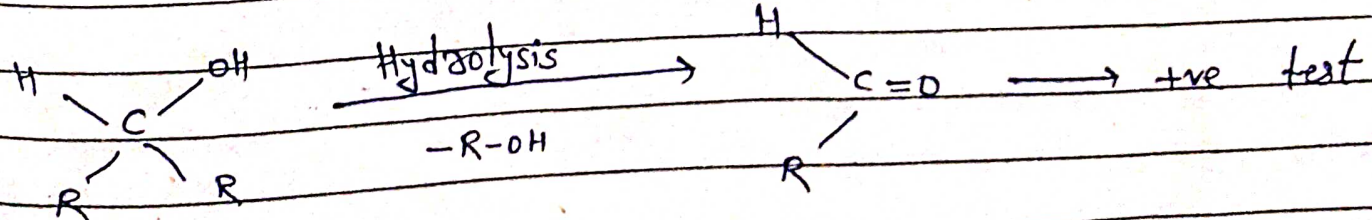
R

give +ve T.R, F.S, B.S Test.



* $H-COOH \xrightarrow{Oxi} CO_2 + H_2O$





14. Test of acidic / active hydrogens {Na metal test}

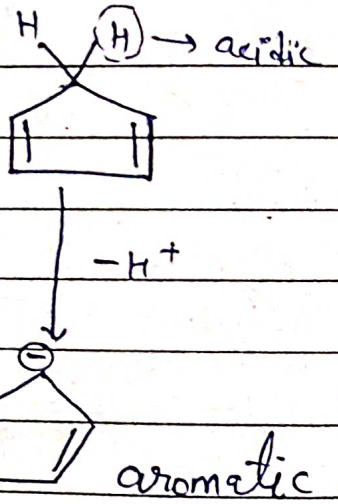
$$Z \rightarrow\!\!\!\!\leftarrow \overset{\delta+}{H} + Na \longrightarrow Na^{+}Z^{-} + \frac{1}{2} H_2 \uparrow$$

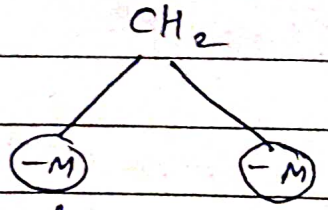
where.

$$Z\text{-}H \Rightarrow \text{—}O\leftarrow \overset{\delta+}{H}\ ,\ -\underset{|}{N}\text{-}H\ ,\ -S\text{-}H\ ,\ \text{-}X\text{-}H$$

$$-C\equiv C\leftarrow \overset{\delta+}{H}\ ,$$

(sp)

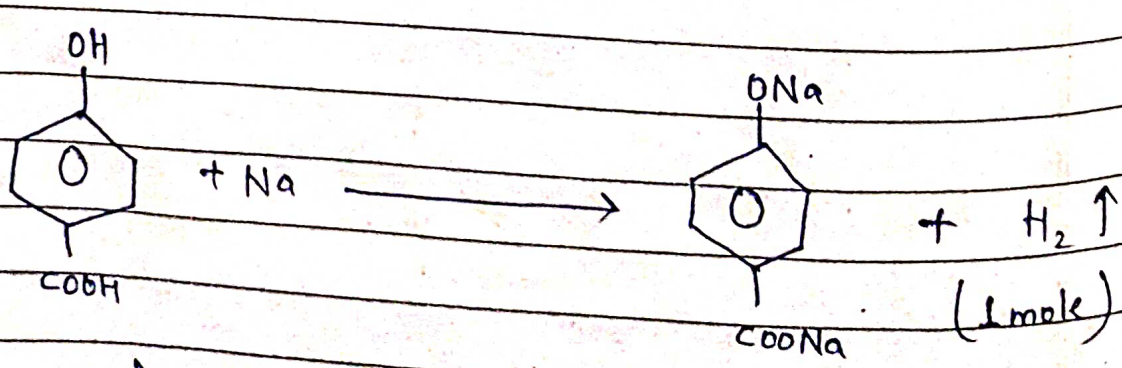




etc

Active methylene group (AMG)

eg :-

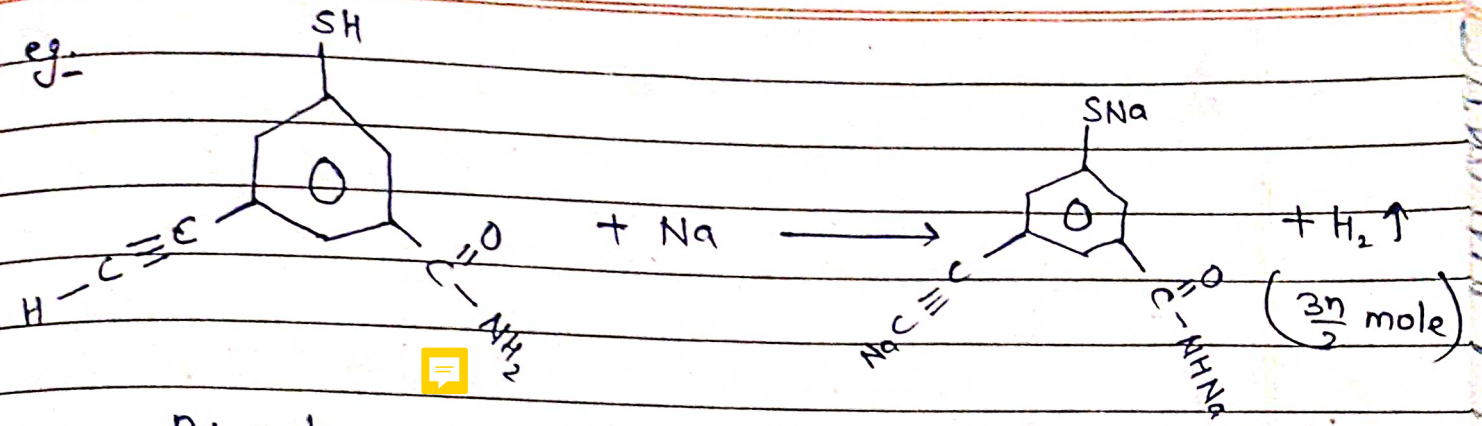


(1 mole)

(1 mole)

Date :

eg:-



n: moles

{3n active H atom}

15. $NaHCO_3$ test.

↳ weak basic solution

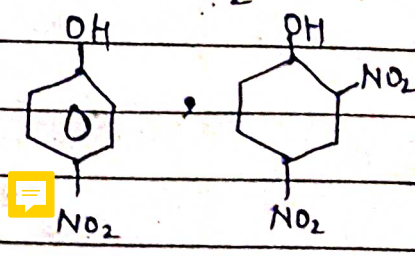
$$A \overset{\delta+}{\leftarrow} H + \overset{+}{Na}\overset{-}{HCO_3} \longrightarrow \overset{+}{A}Na + H_2CO_3 \rightarrow CO_2 + H_2O$$

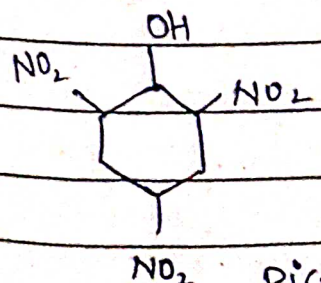
(acid) (unstable)

Condition :- A–H should be stronger acid than H_2CO_3

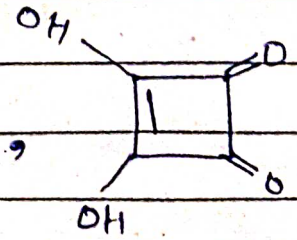
where A–H ⇒ R–COOH , $R-SO_3H$, H_2SO_4 ,

HNO_3 , HCl ,





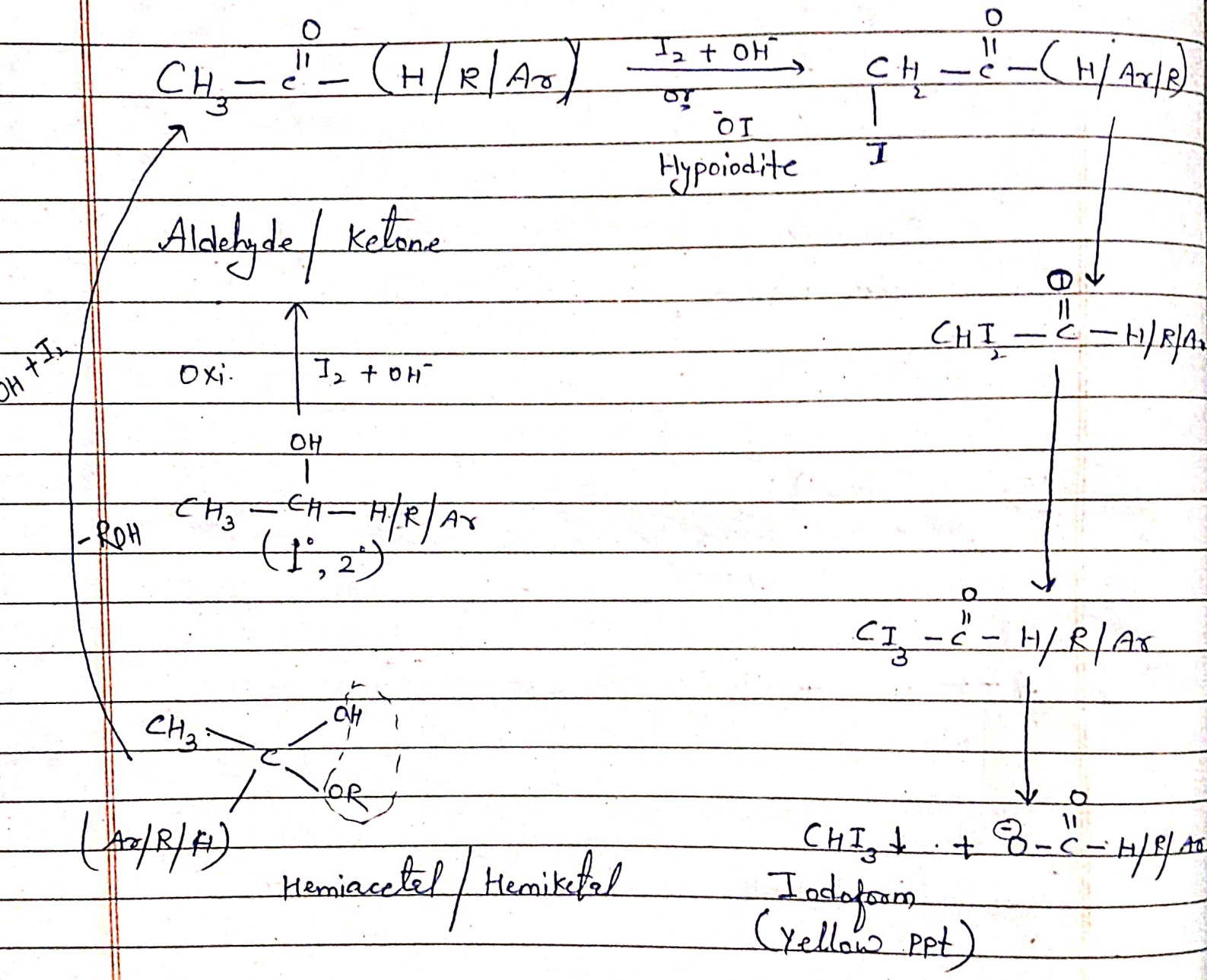
picricacid

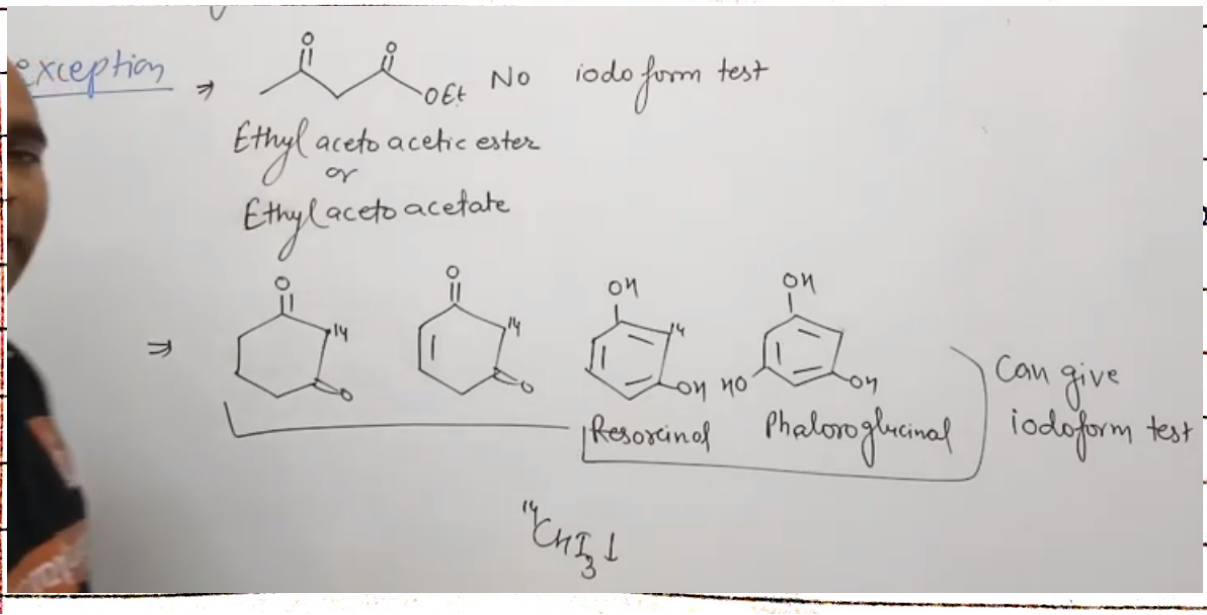


squaric acid.

etc

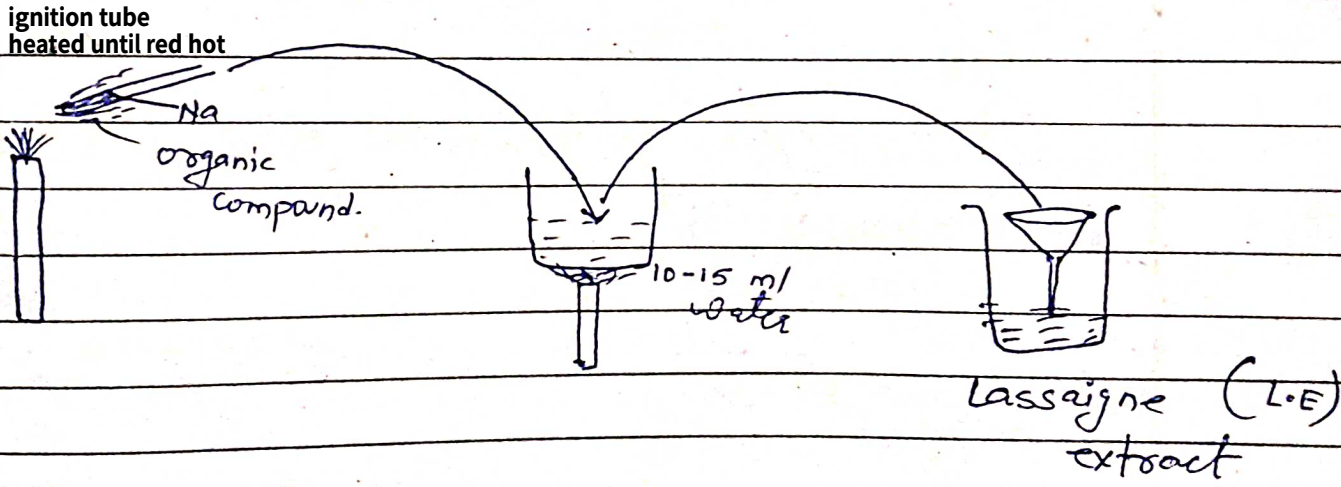
16. Iodoform test.





II Detection of elements :-

- Formation of Lassaigne Extract (L.E.)



Organic Compound $\xrightarrow{Na, \Delta}$ ionic Compound.

① For nitrogen

$$Na + \underbrace{C + N}_{O.C} \xrightarrow{\Delta} NaCN$$

② for S.

$$Na + \underbrace{S}_{O.C} \xrightarrow{\Delta} Na_2S$$

(iii) Halogens

$$Na + X \xrightarrow{\Delta} NaX.$$

(iv) Nitrogen & sulphur

$$Na + S + C + N \longrightarrow NaSCN.$$

(A) Test of Nitrogen:-

$$Na + O.C \xrightarrow{\Delta} \underset{(L.E)}{NaCN} \longrightarrow Na_4[Fe(CN)_6]$$

$$Fe^{+2} \xrightarrow{H_2SO_4} Fe^{+3}$$

Oxidising agent.

$$Na_4[Fe(CN)_6] \xrightarrow{H_2SO_4, \Delta} \text{formed } Fe^{3+}$$

$$Fe^{III}[Fe^{II}(CN)_6]_3$$

Prussian blue complex.

(B) Test of sulpher :-

$$Na + Oc \xrightarrow{\Delta} Na_2S$$

$Na_2S + (CH_3COO^-)_2Pb \rightarrow PbS\downarrow$ Black ppt

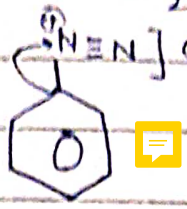
$Na_2S + HgCl_2 \rightarrow HgS\downarrow$ "

$Na_2S + Na_2[Fe(CN)_5(NO)]$ (Sod. Nitroprusside) $\rightarrow Na_4[Fe(CN)_5NOS]$

Purple.

Note:- $NH_2 - NH_2$, $NH_2 - OH$

$Ar - N_2Cl$



NO Lassaigne test of nitrogen.

(c) Test of Nitrogen + Sulphur:-

$$Na + C + S \xrightarrow{\Delta} NaSCN \xrightarrow{FeCl_3} Fe(SCN)_3$$

Sod. Thiocyanate

Blood Red Colour.

PRACTICAL ORGANIC CHEMISTRY

(II) Detection of elements:-

(I) Test of Halogen.

(i) $AgNO_3$ Test

$Na + O.C. \xrightarrow{\Delta} NaX \xrightarrow{HNO_3, \Delta}$

NaX → L.E.

- $AgNO_3$ sol. (if chloride) → AgCl↓ white ppt → NH_4OH sol. → Soluble
- $AgNO_3$ (if Bromide) → AgBr↓ Pale yellow → " → Partially soluble
- $AgNO_3$ (if iodide) → AgI↓ yellow → " → Insoluble

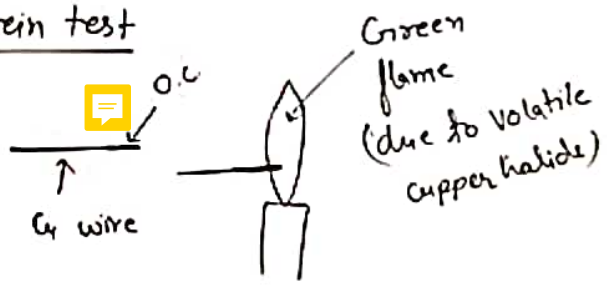
K_{sp}

PRACTICAL ORGANIC CHEMISTRY

(II) Detection of Elements:-

(D) Test of Halogen

(ii) Beilstein test



⇒ Not distinguish to halides. Cannot distinguish between different halides using this test.

PRACTICAL ORGANIC CHEMISTRY

(II) Detection of elements:-

(D) Test of Halogens

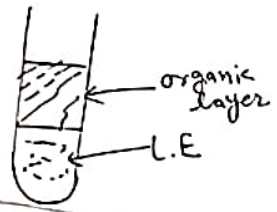
(iii) Organic layer test:- (For Br, I)

$Br^{\ominus} + Cl_2 \longrightarrow Br_2 + Cl^{\ominus}$ (Red-Brown)

$Br^- \xrightarrow{MnO_4^-} Br_2 + MnO_2$

(L.E.)

CS_2 or CCl_4 (Organic solvent)



(Cl_2 water or $KMnO_4$)

Shake

if Br^- → Orange organic layer

(for I^-)

$I^{\ominus} \xrightarrow{KMnO_4} I_2$

$I^{\ominus} + Cl_2 \longrightarrow I_2 + Cl^-$

Violet

→ Violet organic layer

PRACTICAL ORGANIC CHEMISTRY

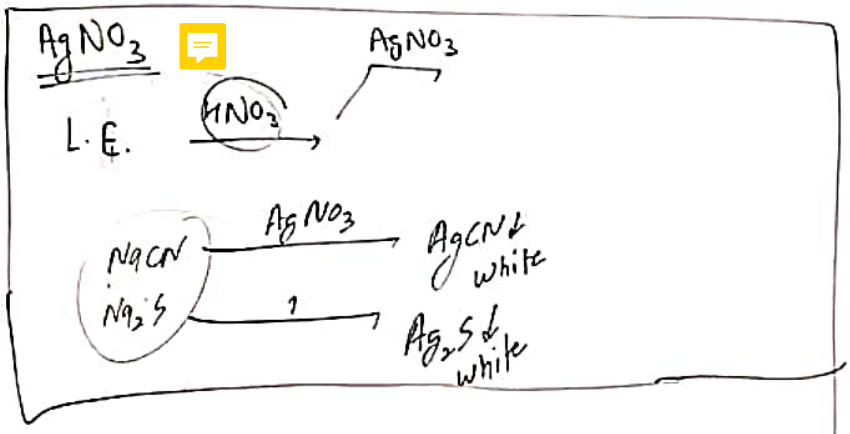
(II) Detection of elements:-

(E) Test of P

O.C. $\xrightarrow[\text{Oxidising agent}]{Na_2O_2}$ Na_3PO_4 $\xrightarrow{HNO_3}$ H_3PO_4 $\xrightarrow{(NH_4)_2MoO_4}$ $(NH_4)_3PO_4 \cdot 12MoO_3$

(P)

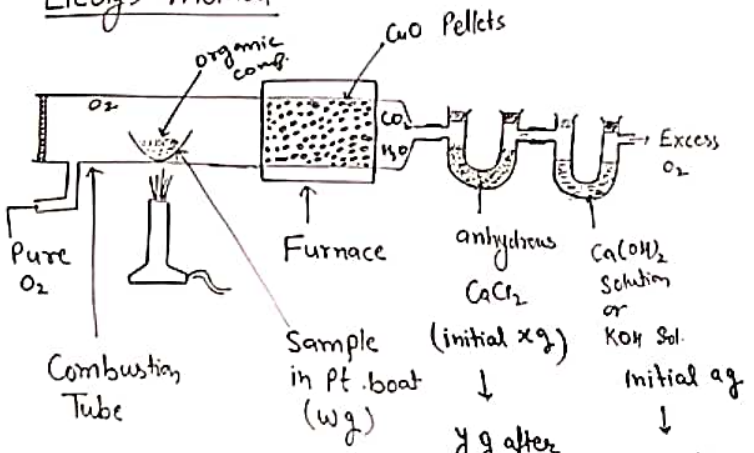
Amm. phospho molybdate (yellow ppt)



QUANTITATIVE ANALYSIS (ESTIMATION OF ELEMENTS)

① Estimation of Carbon and Hydrogen.

Liebig's method



Organic comp. $+O_2 \xrightarrow{\Delta}$ $CO_{2(g)} + H_2O_{(g)} + Cu$ (+ CuO used for oxidation)

- For estimation of C → KOH solution.
- For estimation of H

mass of $CO_2 = (b-a)$ g

$$\%C = \frac{\text{mass of carbon}}{\text{mass of org. Sample (w)}} \times 100$$

$$\%C = \frac{(b-a) \times 12}{44 \times w} \times 100$$

mass of $H_2O = (y-x)$ g

$$\%H = \frac{\text{mass of H}}{\text{Mass of sample (w)}} \times 100$$

$$\%H = \frac{(y-x) \times 2 \times 1}{18 \times w} \times 100$$

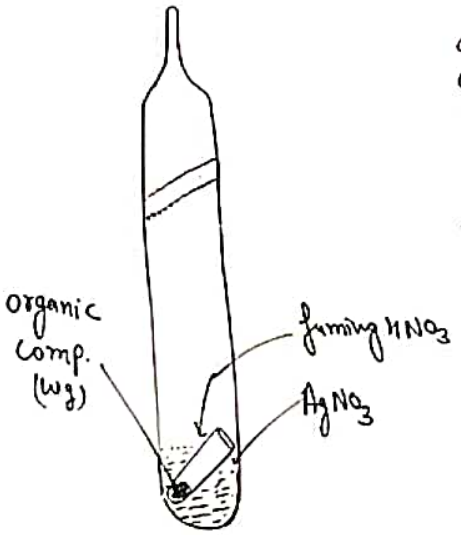
since one molecule of H2O has 2 molecules H, hence multiply by 2

QUANTITATIVE ANALYSIS (ESTIMATION OF ELEMENTS)

(2) Estimation of Halogens:-

Cl = 35.5	
Br = 80	Ag = 108
I = 127	

Carius Method :-



Organic comp. (wg) $\xrightarrow{HNO_3\ \Delta}$ $\xrightarrow{AgNO_3}$ $AgX\downarrow$ xgm

- $X = Cl$ → $AgCl\downarrow$ (x)
- $X = Br$ → $AgBr\downarrow$ (x)
- $X = I$ → $AgI\downarrow$ (x)

$$\Rightarrow \text{For } \%\ Cl = \frac{x \times 35.5}{M_{AgCl} \times w} \times 100$$

$$\Rightarrow \text{For } \%\ Br = \frac{x \times 80}{M_{AgBr} \times w} \times 100$$

$$\Rightarrow \text{For } \%\ I = \frac{x \times 127}{M_{AgI} \times w} \times 100$$

QUANTITATIVE ANALYSIS (ESTIMATION OF ELEMENTS)

(3) Estimation of Sulphur

$$\text{Organic comp } (W\ g) \xrightarrow[\text{(ii) } BaCl_2 \text{ (excess)}]{\text{(i) } HNO_3 \text{ or } Na_2O_2} BaSO_4 \downarrow\ x\ g$$

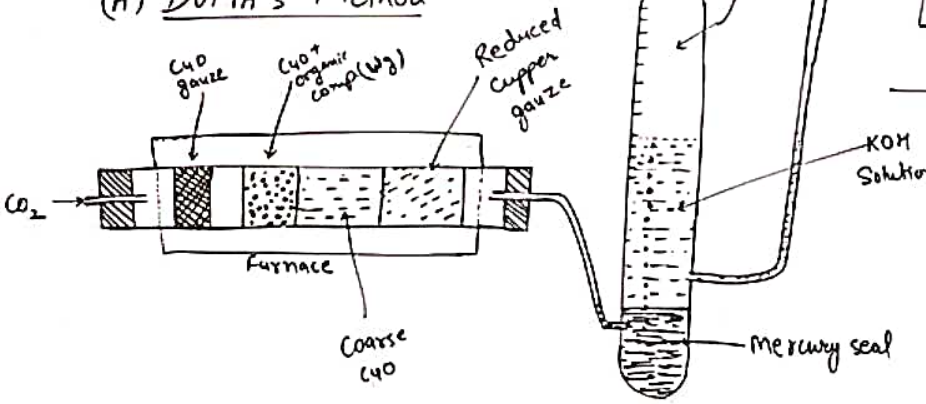
$$\% S = \frac{\text{mass of S}}{\text{Mass of org. Sample}} \times 100$$

$$\% S = \frac{x \times 32}{M_{BaSO_4} \times W} \times 100$$

QUANTITATIVE ANALYSIS (ESTIMATION OF ELEMENTS)

(4) Estimation of Nitrogen

(A) DUMA's Method



$$\%N = \frac{\frac{V_2}{22400} \times 28}{W} \times 100$$

Nitrogen oxides $\xrightarrow{Cu\ \Delta}$ N_2 + CuO

Exp. $\frac{P_1 V_1}{T_1}$ = ideal (STP) $\frac{P_2 V_2}{T_2}$

$$V_2 = \frac{P_1 V_1}{T_1} \times \frac{T_2}{P_2}$$

$T_2 \leftarrow 273$ K

$P_2 \leftarrow 760$ mm of Hg

V_2 = Vol. (ml) of N_2 in ideal condition

$P_{atm} = P_{N_2} + P_{H_2O}$

$P_{N_2} = P_{atm} - P_{H_2O}$

P_1 = aq. tension

$$n = \frac{PV}{RT}$$

Organic comp.

$C_x H_y N_z O + CuO \xrightarrow{\Delta} CO_2 + H_2O$

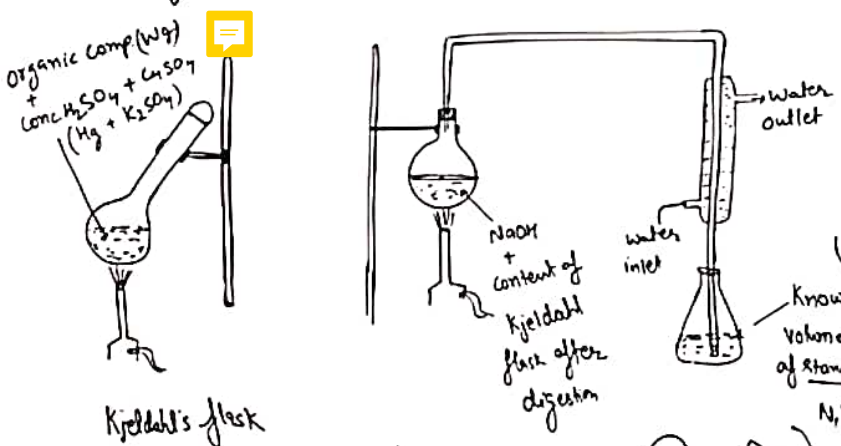
N_2 + Nitrogen oxide + Cu ↓

Nitrogen oxides $\xrightarrow{Cu\ \Delta}$ N_2 + CuO

QUANTITATIVE ANALYSIS (ESTIMATION OF ELEMENTS)

(4) Estimation of Nitrogen

(B) Kjeldahl's Method :



→ in Kjeldahl's flask

$$C + H + N + S \xrightarrow[(CuSO_4/Hg) + K_2SO_4]{H_2SO_4\ \Delta} CO_2 + H_2O \quad SO_2 + NH_3$$

$$NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$$

$$\Rightarrow NH_4SO_4 + NaOH \longrightarrow Na_2SO_4 + NH_3\uparrow$$

(in flask) H_2SO_4 (N_1V_1) + $2NH_3$ → $(NH_4)_2SO_4$ + (H_2SO_4 remaining)

(excess)

Titration: $NaOH$ (N_2V_2) — End point

Note : Not useful for $-NO_2$, $-N=N-$, N in ring,

since these do not form NH_3 in reaction conditions

QUANTITATIVE ANALYSIS (ESTIMATION OF ELEMENTS)

(4) Estimation of Nitrogen

(B) Kjeldahl's Method :

Titration

$2NH_3 + H_2SO_4 \longrightarrow (NH_4)SO_4 + H_2SO_4$ (remaining)

H_2SO_4: M_1V_1, ↑ molarity

remaining H_2SO_4 ↓ titrate with NaOH (M_2V_2) → end

$H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4$

remaining: M_2V_2

moles of H_2SO_4 (remaining) $= \frac{\text{mole NaOH}}{2} = \frac{M_2V_2}{2}$

initial moles of $H_2SO_4 = M_1V_1$

Reacted moles of H_2SO_4 (with NH_3) $= \left[M_1V_1 - \frac{M_2V_2}{2}\right]$

mole of $NH_3 = 2\left(M_1V_1 - \frac{M_2V_2}{2}\right)$

$$\text{Molarity \% N} = \frac{2\left(M_1V_1 - \frac{M_2V_2}{2}\right) \times 14}{W} \times 100$$

QUANTITATIVE ANALYSIS (ESTIMATION OF ELEMENTS)

Q. 0.30 gm of an organic compound gave 50ml of nitrogen gas collected at 300K and 715 mm of Hg pressure in Dumas method. Calculate the % of nitrogen in compound (aqueous tension of water at 300K is 15 mm of Hg.)

Sol.

Experimental condition

$P_1 = (P_{atm} - P_{H_2O}) = 715 - 15 = 700$ mm of Hg

$V_1 = 50$ ml

$T_1 = 300$K

STP

$P_2 = 760$

$V_2 = ?$

$T_2 = 273$K

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\frac{700 \times 50}{300} = \frac{760 \times V_2}{273}$$

$$V_2 = \frac{700 \times 50 \times 273}{300 \times 760} = 41.9 \text{ ml}$$

Vol. N_2 (STP)

$$\% N = \frac{41.9 \times 28}{22400 \times 0.3} \times 100 = 17.45\%$$

QUANTITATIVE ANALYSIS (ESTIMATION OF ELEMENTS)

Q 0.50 gm of an organic compound was treated according to Kjeldahl's method. The evolved ammonia was absorbed in 50ml of 0.5M H_2SO_4 solution. The residual acid required 60 ml of $\frac{M}{2}$ NaOH solution. Calculate % of Nitrogen in comp.

Sol $2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4 + H_2SO_4$ (remaining)

20 mmoles

H_2SO_4: 0.5 M, 50 ml, 25 mmoles

remaining H_2SO_4: 15 mmoles

NaOH (0.5M), 60ml → end point

$2NaOH + H_2SO_4 \longrightarrow Na_2SO_4$

60ml, 0.5 M, 30 mmoles

H_2SO_4: 15 mmoles

reacted mmoles of H_2SO_4 = 25 - 15 = 10 mmoles

$2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$

20 mmoles, 10 mmoles

$$\%N = \frac{20 \times \quad \times 14}{0.5 \times 1000} \times 100 = 56\%$$

QUANTITATIVE ANALYSIS (ESTIMATION OF ELEMENTS)

Q 0.15 gm of an organic substance when treated by Carius method gave 0.12 g of Silver bromide. Calculate percentage of bromine in the compound. (At. wt Ag = 108, At. wt Br = 80)

Sol Org. comp. (Br) ⟶ ⟶ ⟶ AgBr↓

0.15 gm = 0.12 g = $\frac{0.12}{188}$

AgBr = M.wt = 108 + 80 = 188

$$\% Br = \frac{0.12}{188} \times \frac{80}{0.15} \times 100 = 34.04\%$$

QUANTITATIVE ANALYSIS (ESTIMATION OF ELEMENTS)

Q 0.2595 gm of an organic compound when treated by carius method gave 0.35g of $BaSO_4$. Calculate the percentage of sulphur in the compound.

At wt. Ba = 137
" S = 32
" O = 16

Sol. Org. comp. (0.2595 gm) ⟶ ⟶ $BaSO_4\downarrow$ (0.35 g)

M.wt of $BaSO_4$

137
96
233

$$\%S = \frac{\text{mass of sulphur}}{\text{mass of org. comp}} \times 100 = \frac{0.35 \times 32}{233 \times 0.2595} \times 100$$

$$= 18.52\%$$

PURIFICATION OF ORGANIC COMPOUNDS

competishun

Removal of undesirable impurities present in sample of compounds is called purification.

Methods of purification.

(I) Sublimation

(II) Crystallisation

(III) Distillation

(IV) Differential extraction

(V) Chromatography

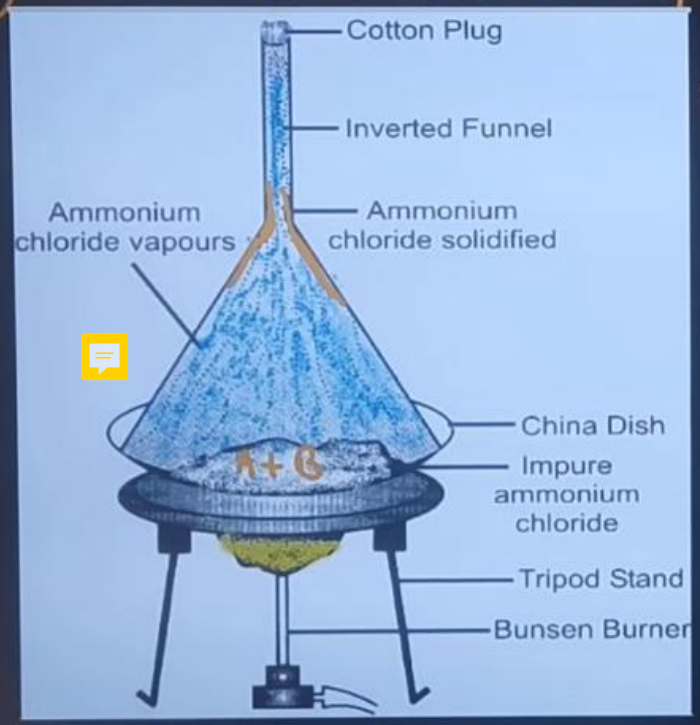
Pure compounds have sharp melting and boiling points. So purity of organic compounds is checked by determination of value of melting and boiling point.

PURIFICATION OF ORGANIC COMPOUNDS

competishun

(I) Sublimation purification :-

This method is used to separate sublimable compounds from nonsublimable impurities.



⇒ Sublimation :- Solid ⟶ gas

⇒ $(A_s + B_s) \xrightarrow{\Delta} A_g^{\uparrow} + B_s$

Sub. Non Sub.

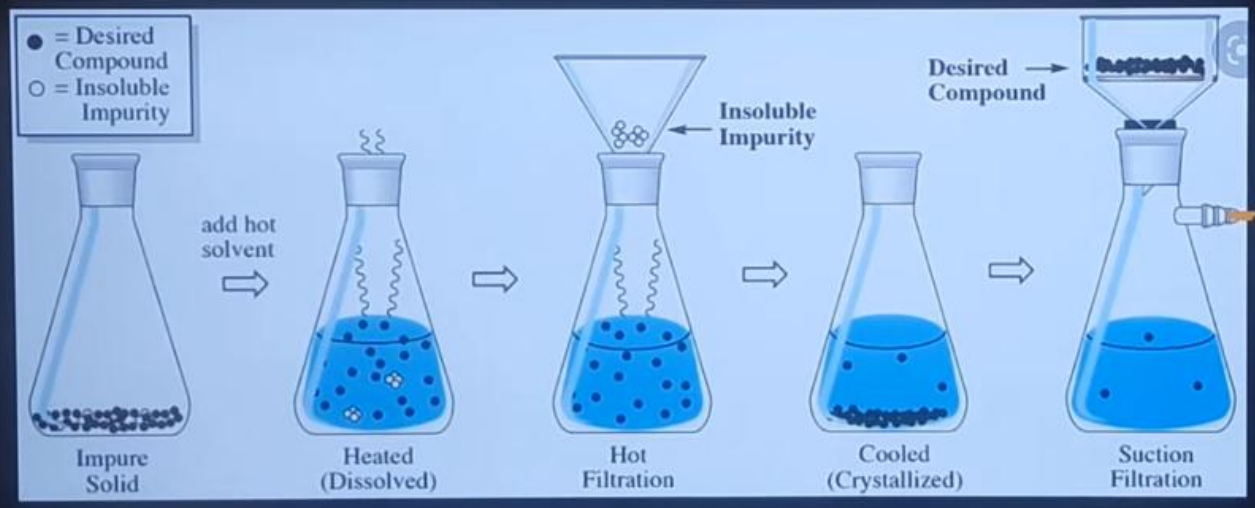
use full = NH_4Cl, Naphthalene, Camphor, iodine

PURIFICATION OF ORGANIC COMPOUNDS

competishun

(II) Crystallition purification :-

This method is based on the difference in solubilities of the compound and impurity in suitable solvent.



eg. Benzoic acid

- insoluble in cold water
- Soluble in hot water

PURIFICATION OF ORGANIC COMPOUNDS

competishun

Removal of undesirable impurities present in sample of nds is called purification.

Methods of purification.

(I) Sublimation
(II) Crystallisation
} For solids.

(III) Distillation
(IV) Differential extraction
} For liq.

(V) Chromatography

Pure compounds have sharp melting and boiling points. So purity of organic compounds is checked by determination of value of melting and boiling point.

PURIFICATION OF ORGANIC COMPOUNDS

competishun

(III) Distillation:-

- (A) Simple distillation
- (B) Fractional distillation
- (C) Distillation under reduced pressure
- (D) Steam distillation

(A) Simple distillation :-

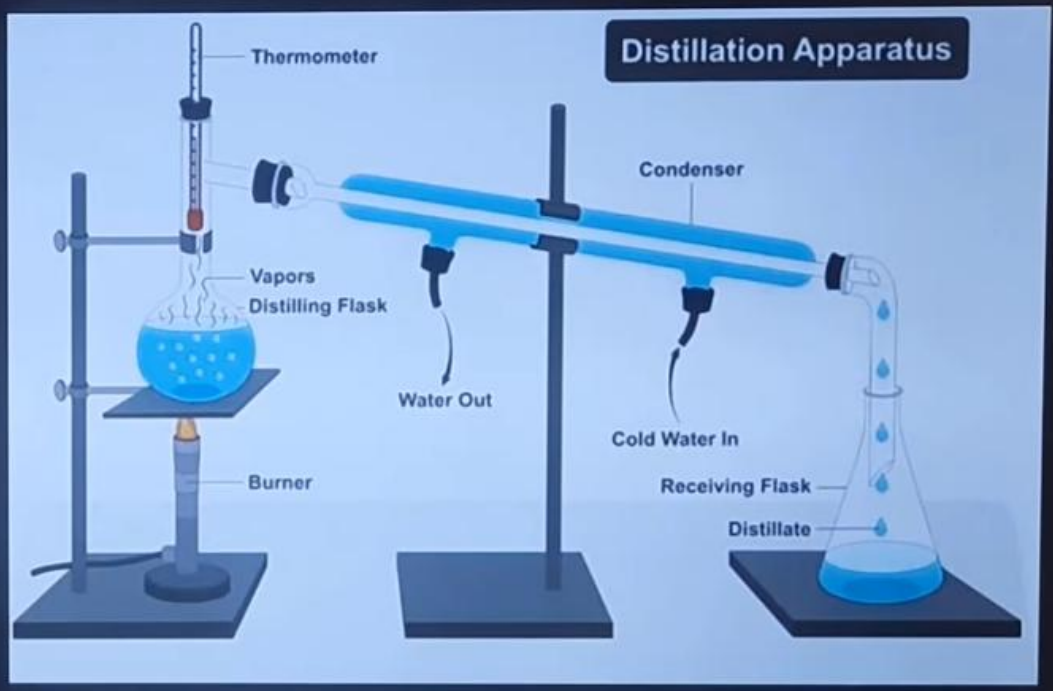
This method is used to separate volatile liquids from non-volatile impurities having sufficient difference in their boiling points.

PURIFICATION OF ORGANIC COMPOUNDS

competishun

eg ($CHCl_3$ + $Ph-NH_2$)

B.P	BP
60°C	189°C



Simple Distillation apparatus

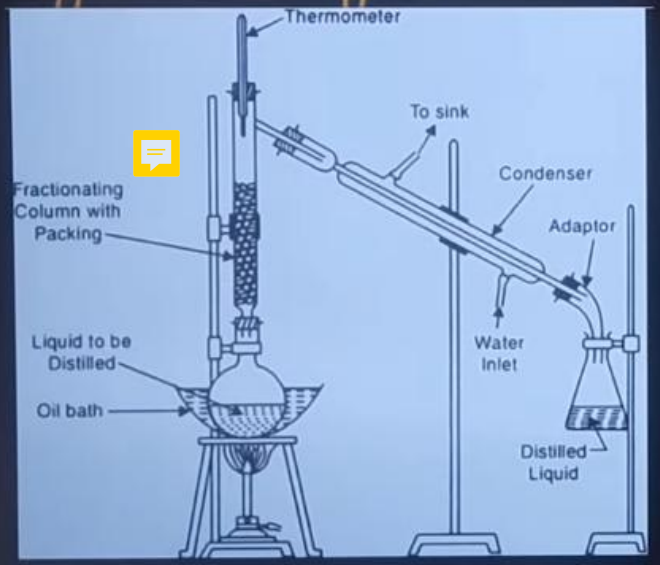
PURIFICATION OF ORGANIC COMPOUNDS

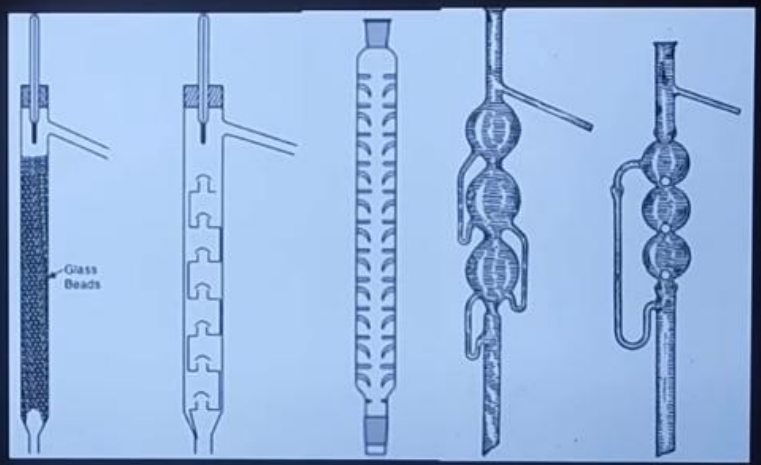
competishun

(B) Fractional distillation

This method is used to separate two liquids when the difference in their boiling points is not much.

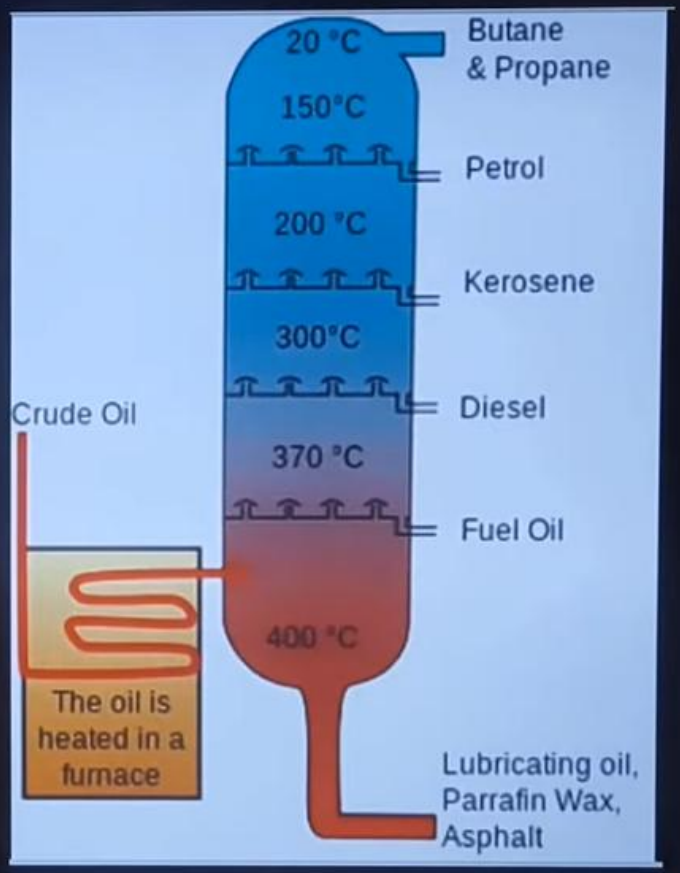
In this method, vapours of the liquid mixture are passed through a fractionating column which is fitted over the mouth of R.B. Flask





PURIFICATION OF ORGANIC COMPOUNDS

competishun



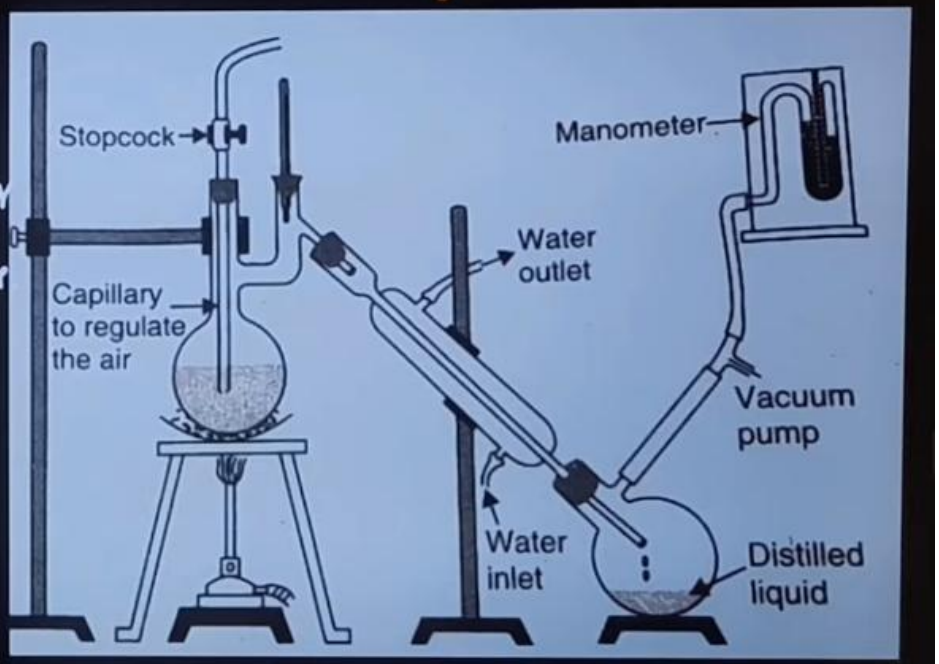
PURIFICATION OF ORGANIC COMPOUNDS

competishun

(C) Distillation under reduced pressure :-

This technique is used to purify liquids having very high boiling points and those which decompose at or below thier boiling points.

The boiling point of a liquid is difined as the temperature at which its vapaour pressure is equal to the external pressur

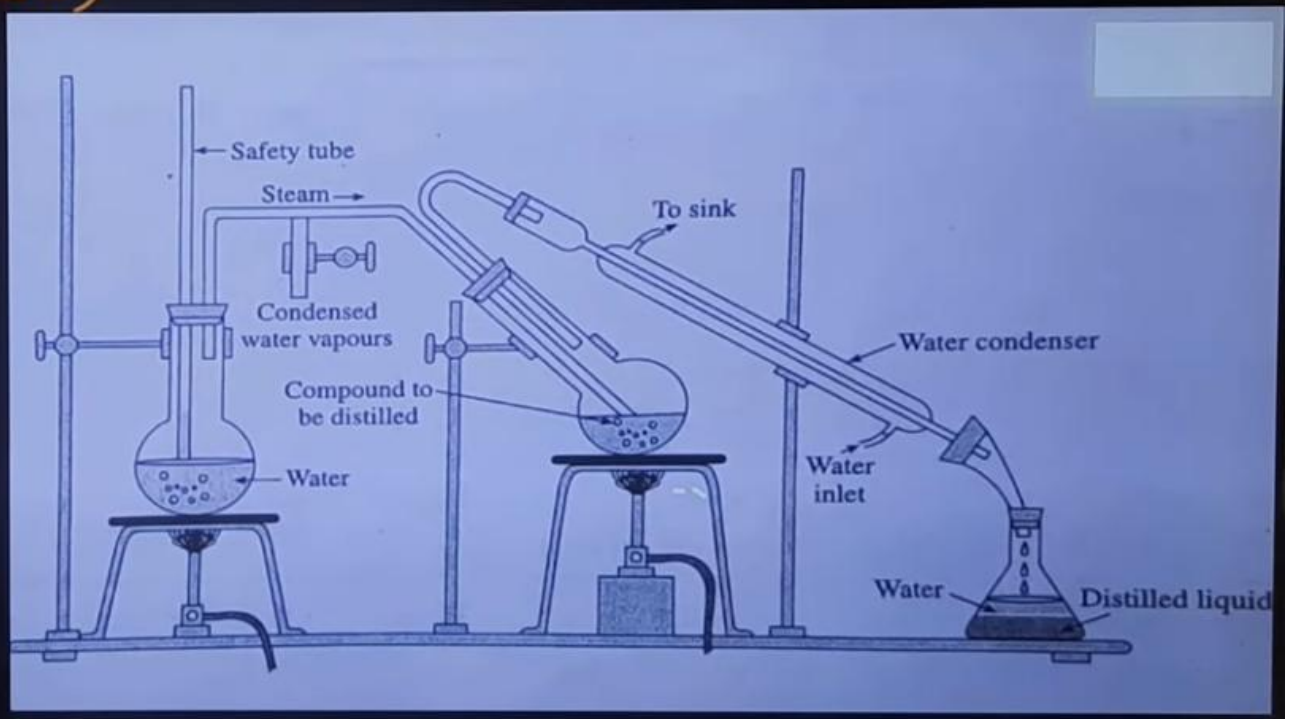


Example :- Separation of glycerol from spent lye in soap industry.

PURIFICATION OF ORGANIC COMPOUNDS

competishun

(D) Steam distillation:- This method is used to separate the compounds are steam volatile and insoluble in water. (like aniline water mixture.)

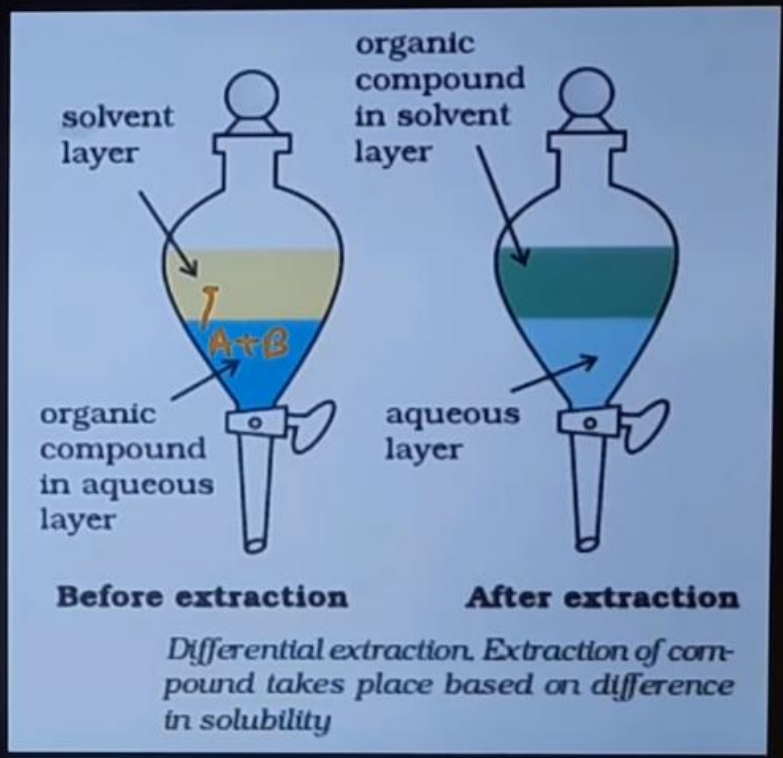


PURIFICATION OF ORGANIC COMPOUNDS

competishun

(IV) Differential extraction :-

This method is based on the different solubility of an organic comp. in different solvents.



Differential extraction. Extraction of compound takes place based on difference in solubility

PURIFICATION OF ORGANIC COMPOUNDS

competishun

(V) Chromatography :-

- Column Chromatography
- Thin layer chromatography (TLC)
- Paper chromatography
- Gas chromatography

⇒ It is based on the principle that different constituents are adsorbed on an adsorbsent to different degrees.

⇒ Stationary phase = adsorbent (Silica gel, activated charcoal)

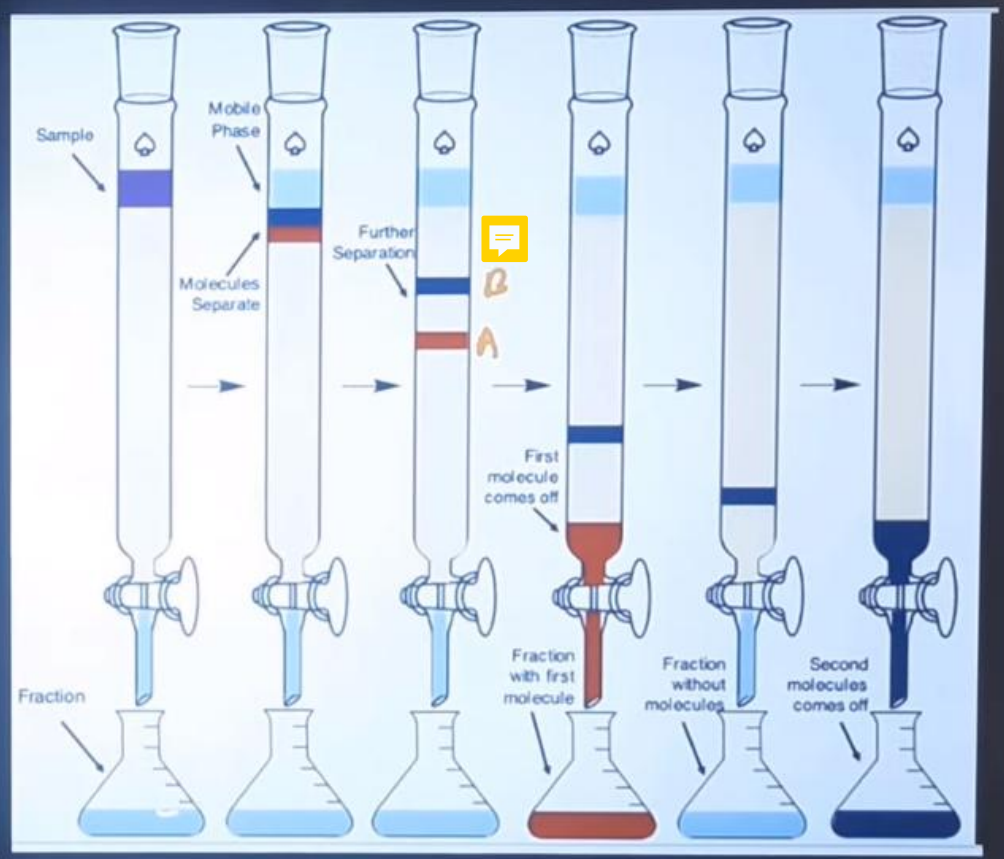
⇒ Moving phase = Mobile phase = (Solvent + component)
↑ (eluant)

PURIFICATION OF ORGANIC COMPOUNDS

competishun

Column Chromatography :-

A + B

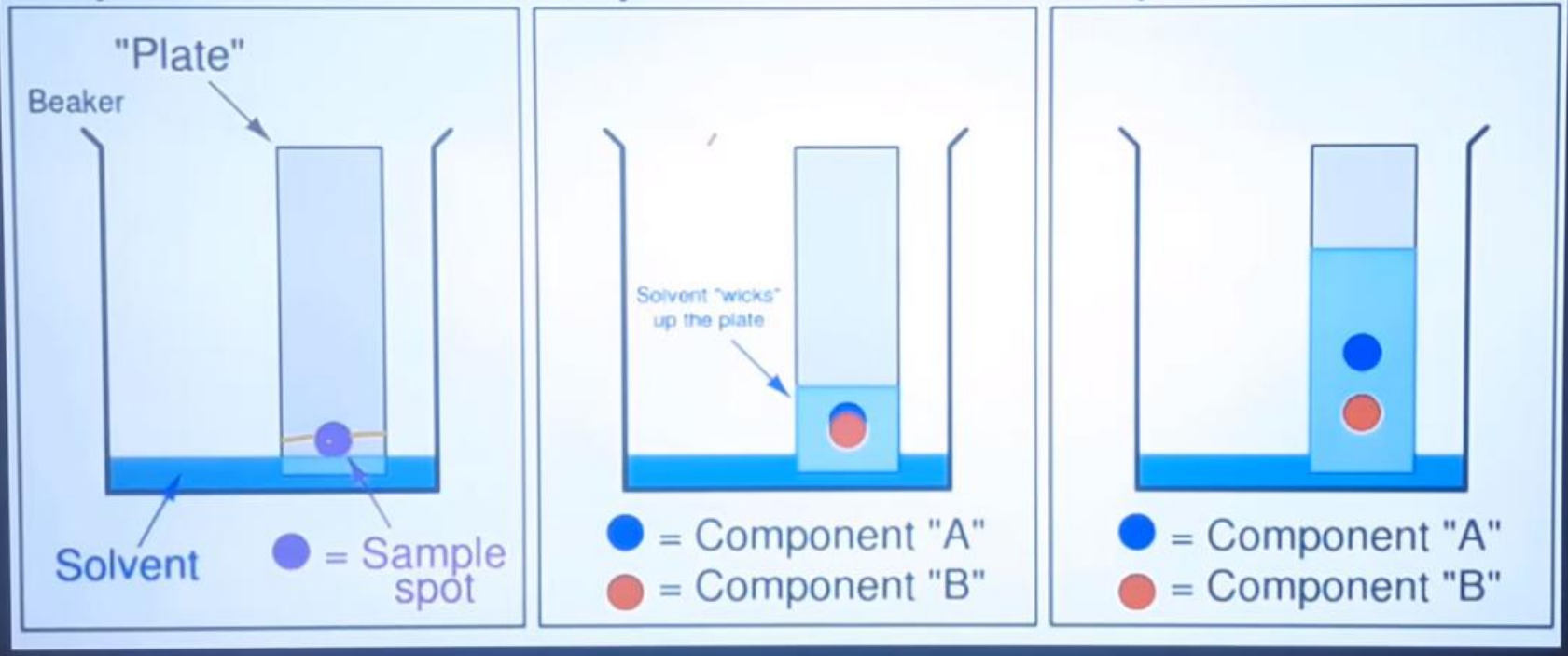


PURIFICATION OF ORGANIC COMPOUNDS

competishun

Thin-layer chromatography

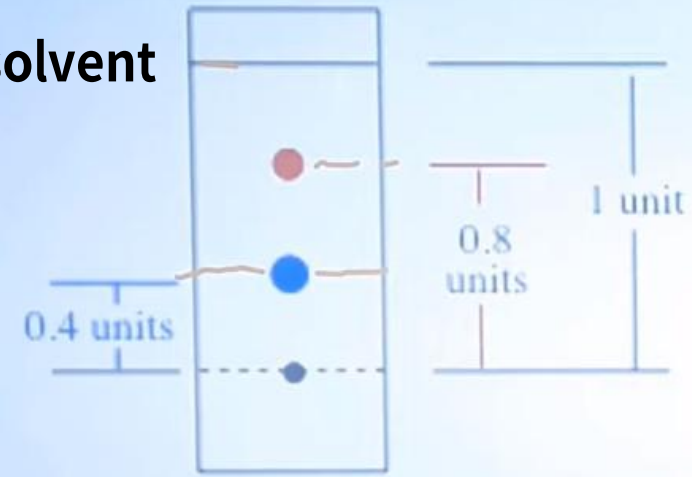
TLC



PURIFICATION OF ORGANIC COMPOUNDS

competishun

Determining Rf Values:



$$Rf = \frac{\text{distance molecule traveled}}{\text{distance solvent traveled}}$$

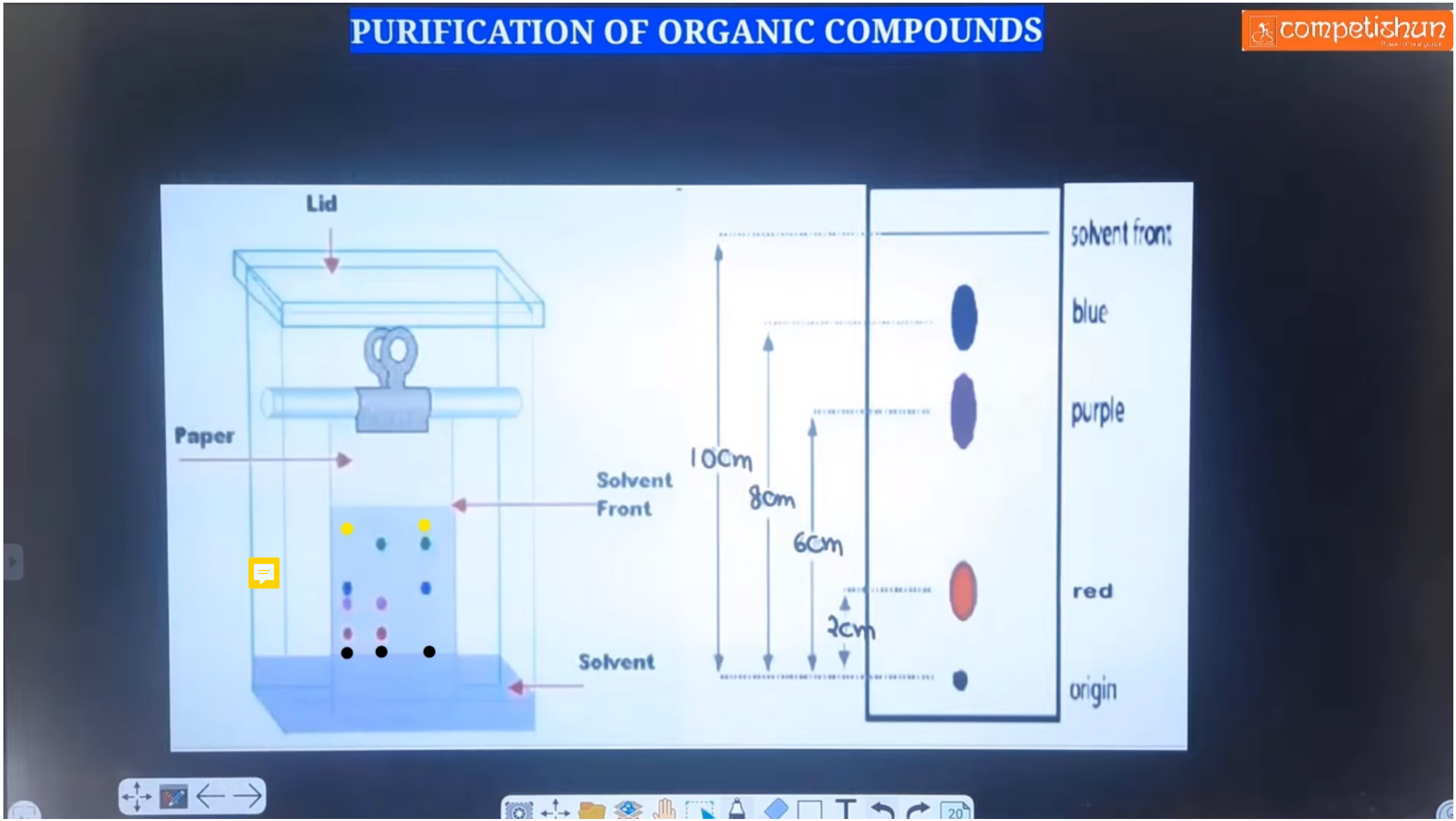
$Rf = 0.4$

$Rf = 0.8$

PURIFICATION OF ORGANIC COMPOUNDS

- R_f value is constant for each component only under identical experimental condition.
- Polar compounds have low R_f value
- It depend on following factors-
  - ✓ Nature of adsorbent
  - ✓ Mobile phase
  - ✓ Activity
  - ✓ Thickness of layer
  - ✓ The temperature
  - ✓ Equilibration
  - ✓ Loading
  - ✓ Dipping zone
  - ✓ Chromatographic technique

competishun



Gas chromatography : chromatography of gases involving similar experimental setup